

**UNITED STATES DISTRICT COURT
FOR THE DISTRICT COURT OF DELAWARE**

QORVO, INC.

Plaintiff,

v.

AKOUSTIS TECHNOLOGIES, INC. and
AKOUSTIS, INC.

Defendants.

C.A. No. 21-1417 (JPM)

JURY TRIAL DEMANDED

THE AKOUSTIS DEFENDANTS' SECOND AMENDED TRIAL EXHIBIT LIST

Dated: May 12, 2024

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*Attorneys for Defendants Akoustis
Technologies, Inc. and Akoustis, Inc.*

Defendants Akoustis Technologies, Inc. and Akoustis, Inc. (collectively "Akoustis")

hereby provide the Amended Exhibit List for trial:

EXHIBIT E.2 – AKOUSTIS' AMENDED EXHIBIT LIST

Dx	Beginning Bates	End Bates	Description
1	QORVO_00844571	QORVO_00844572	Email from Robert Aigner to Mike Fresina dated 5/18/2015 re: "Qorvo-Akoustis NDA and Meeting"
2	AKTS_00011800	AKTS_00011814	Email from Robert Aigner to David Aichele dated 6/14/2015
3	AKTS_00011815	AKTS_00011834	Email from Dave Aichele to Robert Aigner dated 6/5/15 re: papers
4	AKTS_00111693	AKTS_00111685	Email from Dave Aichele to Rama Vetury dated 11/30/15 re: papers
5	AKTS_00070112	AKTS_00070115	Email from Robert Aigner to Dave Aichele dated 12/1/15 re: papers
6	QORVO_00844533	QORVO_00844533	Email from Dave Aichele to Robert Aigner dated 12/31/15 re: RWW Paper
7	QORVO_00868286	QORVO_00868312	Email from Dave Aichele to Todd Gillenwater, et al re Qorvo Visit, with attached RFIC2016
8	AKTS_00115640	AKTS_00115661	Email from Robert Aigner to Rama Vetury dated 7/15/16 re: follow up, presentation attached, with attached "Recent developments in AIScN thin film investigations" presentation
9	QORVO_00844608	QORVO_00844608	Email from Robert Aigner to Berry Leonard dated 11/2/16 re: http://www.akoustis.com/disruptive-single-crystal-filter-solutions-for-4glte-smartphones-and-beyond/
10	QORVO_00844610	QORVO_00844619	Email from Robert Aigner to Berry Leonard dated 11/3/16 re: http://www.akoustis.com/disruptive-single-crystal-filter-solutions-for-4glte-smartphones-and-beyond/ , with attached Response to Akoustis rev2.pptx
11	AKTS_00075254	AKTS_00075266	Email from Rohan Houden to Rama Vetury dated 6/2/16 re: aigner follow up

DX	Beginning Bates	End Bates	Description
12	AKTS_00033980	AKTS_00033987	Email from Robert Aigner to Rama Vetury dated 1/19/17 re: follow up, presentation attached, with attached pages from "Piezoelectric MEMS Resonators"
13	AKTS_00117354	AKTS_00117357	Email from Rama Vetury to Dave Aichele dated 2/1/17 re: BAW press release
14	QORVO_00079137	QORVO_00079224	2/21/17 Agenda: Acoustic TSC8
15	QORVO_00844185	QORVO_00844189	Email from Robert Aigner to Todd Gillenwater dated 7/24/17 re: Akoustis paper IMS2017
16	QORVO_00867118	QORVO_00867123	Email from Todd Gillenwater to Eric Creviston dated 7/31/17 re: Filter briefing for analyst meetings, DRAFT version, with attached Q&A about the competitive landscape in BAW and SA
17	QORVO_00844060	QORVO_00844061	Email from Steve Grant to Robert Aigner and Otto Berger dated 11/20/17 re: Akoustic Receives First Purchase Order From A Top-5 Smartphone OEM
18	QORVO_00888308	QORVO_00888424	CEO and CFO Ops Review Fab Technology and Manufacturing, December 2017
19	QORVO_00844094	QORVO_00844096	Email from Bob Bruggeworth to Brian Balut dated 5/12/18 re: Akoustis Announces New Tier-One Mobile Infrastructure QEM Customer and Purchase Orders for RF Filters
20	QORVO_00844086	QORVO_00844093	Email from Alex Zajac to Brian Balut dated 5/22/18 re: Akoustis at CSMantech, with attached "5.2 GHz Rnd Test Filter 20180220. pdf
21	QORVO_00079087	QORVO_00079101	Qorvo Presentation dated 2018 titled "Objective: New Product/Process Development Speed"
22	QORVO_00844325	QORVO_00844325	Email from Bob Sowell to Ken Schroeter dated 11/1/28 re: Call Follow-Up
23	QORVO_00844620	QORVO_00844624	Email from Robert Aigner to Gernot Fattinger dated 11/5/18 re: Akoustis Test Plan DRAFT, with attached High Power, Wideband Single Crystal XBAW

DX	Beginning Bates	End Bates	Description
			Technology for sub-6 GHz Micro RF Filter Applications article
24	QORVO_00866372	QORVO_00866460	Email from Todd Gillenwater to Robert Aigner dated 3/20/19 re: Akoustis Data, with attached 5.25GHz Filter presentation
25	QORVO_00844204	QORVO_00844204	Email from Robert Aigner to Todd Gillenwater dated 5/9/19 re: Oppenheimer note on AKTS mentions QRVO twice
26	QORVO_00844730	QORVO_00844738	Email from Robert Aigner to Ben Thomas dated 5/31/19 re: Oppenheimer note on AKTS mentions QRVO twice, with attached "BAW filters for 5G: Facts vs. Fiction"
27	QORVO_00039748	QORVO_00039786	Email from Robert Aigner to Marck Solal and Alireza Tajic dated 9/10/19 re: Microwave Journal Filter Article book, with attached 5G Filtering Solutions eBook
28	AKTS_00282924	AKTS_00282925	Email from Rohan Houlden to Jeffrey Shealy dated 5/7/20 re: feedback from Aigner
29	QORVO_00635103	QORVO_00635107	Robert Aigner's Qorvo Confidentiality, Non-Solicitation and Non-Competition Agreement, dated 4/15/21
30	QORVO_01200133	QORVO_01200136	Article - <i>Corrective Actions to Meet Extreme Tolerance Requirements for Thin Fims</i>
31	AKTS_00755868	AKTS_00756159	Article - <i>RF Bulk Acoustic Wave Filters for Communications</i>
32	N/A	N/A	Declaration of Robert Aigner, Ph.D. in Support of Triquint's Motion for Partial Summary Judgment
33		N/A	Intentionally Omitted
34	N/A	N/A	Declaration of Robert Aigner, Ph.D. in Support of Triquint's Motion to Seal
35			Intentionally Omitted
36			Intentionally Omitted
37			Intentionally Omitted

DX	Beginning Bates	End Bates	Description
38	QORVO_00888251	QORVO_00888267	FCI License Agreement, dated 4/5/07
39	QORVO_00888178	QORVO_00888189	Integrated Inductor Lead Frame License Agreement between RFMD and AMKOR, dated 1/1/03
40	QORVO_00888207	QORVO_00888231	Patent License Agreement between Fujitsu Media Devices Limited and Sawtek Incorporated, dated March, 2003
41	QORVO_00888190	QORVO_00888206	Patent License Agreement between Toshiba and TriQuint dated 3/20/14
42			Intentionally Omitted
43			Intentionally Omitted
44			Intentionally Omitted
45			Intentionally Omitted
46			Intentionally Omitted
47			Intentionally Omitted
48			Intentionally Omitted
49			Intentionally Omitted
50			Intentionally Omitted
51			Intentionally Omitted
52			Intentionally Omitted
53			Intentionally Omitted
54			Intentionally Omitted
55	QORVO_00689611	QORVO_00689627	Email from Tony Testa to John Anderton and Wayne Polonio dated 8/2/19 re: QPQ1904 5.6GHz Design Review, with attached QPQ1904 (uBAW, 8") 5.6 GHz WiFi Filter Design
56	AKTS_00082511	AKTS_00082515	Email from Jeffrey Shealy to Dave Aichele and Rohan Houlden dated 10/2/18 re: Akoustis_WCON Spec. pptx

DX	Beginning Bates	End Bates	Description
57	QORVO_00844666	QORVO_00844670	Email from Robert Aigner to Gernot Fattinger dated 5/29/2016 re: RFIC paper, with attached article, <i>Single Crystal AlGaIn Bulk Acoustic Wave Resonators on Silicon Substrates with High Electromechanical Coupling</i>
58	QORVO_00844564	QORVO_00844563	AKF-1252 Evaluation Samples, September 2017
59	QORVO_00886071	QORVO_00886074	Email from Leah Giovan to Todd Gillenwater dated 11/7/18 re: Microwave Journal Article showcasing 5GHz BAW from Akoustis, with attached Qorvo 5.2GHz vs. Akoustis
60	QORVO_00079389	QORVO_00079413	Presentation - Filter Technology Update
61	AKTS_00040093	AKTS_00040098	Article - <i>Pushing BAW beyond 'known' Frontiers" Higher, Wider, Smaller, Cooler</i>
62			Intentionally Omitted
63	AKTS_00114906	AKTS_00114909	Article - Optimization of Acoustic Dispersion for High Performance Thin Film BAW Resonators
64			Intentionally Omitted
65			Intentionally Omitted
66			Intentionally Omitted
67	QORVO_00305360	QORVO_00305385	Introduction to PDE and Document Center dated 1/3/15
68	QORVO_00718915	QORVO_00719061	Email from Tony Testa to Jerry Gray, Ron Sanov, Billy Elliott, Bob Baeen, Jeff Cameronk, and Ahad Abdelmajid dated 1/2/15 re: FW: Product Training Webinars for Sales, Presentation by Day 30
69	QORVO_00844058	QORVO_00844058	Symantec Invoice Number 200430666 dated 6/1/16 to Triquint Semiconductor Inc.
70	QORVO_00843661	QORVO_00843666	Symantec Rewards Program Agreement between Symantec and TriQuint Semiconductor dated 3/27/2008
71	QORVO_00843667	QORVO_00843667	Symantec Rewards Territory Agreement Addendum between Symantec and TriQuint Semiconductor dated 3/27/2008

DX	Beginning Bates	End Bates	Description
72	QORVO_00843681	QORVO_00843683	Symantec General Non-Disclosure Agreement between Symantec Corporation and Qorvo, Inc. dated 8/12/16
73	N/A	N/A	Declaration of Michael Boyd dated 5/19/23 in support of Qorvo's Answering Brief in Opposition to Defendants Motion to Re-Designate the Second Amended Complaint Marked Attorneys' Eyes Only to Confidential
74	N/A	N/A	Qorvo's Third Supplemental Objections and Responses to Akoustis' Second and Third Set of Interrogatories (Nos. 10-12) dated 9/26/23
75	QORVO_00842104	QORVO_00842112	Qorvo's Policy Statement for data creation and data security dated 3/21/2019
76	QORVO_00842868	QORVO_00842873	Qorvo's Policy Statement for data creation and data security dated 3/29/2021
77	QORVO_00842856	QORVO_00842867	Qorvo's End User Data Classification dated 7/14/2023
78	QORVO_00842846	QORVO_00842851	Using Bolden James Classifier - Data Classification for Windows dated 5/14/2023
79	QORVO_00843768	QORVO_00843771	Digital Guardian Subscription Agreement with Qorvo dated 2/1/2020
80	QORVO_00072622	QORVO_00072637	Qorvo Audit Committee Update Information Technology dated 2/5/2019
81	QORVO_00072556	QORVO_00072574	Qorvo Audit Committee Update Information Technology dated 5/9/2019
82	QORVO_00072592	QORVO_00072608	Qorvo Audit Committee Update Information Technology dated 8/6/2019
83	QORVO_00072575	QORVO_00072591	Qorvo Audit Committee Update Information Technology dated 10/28/2019
84	QORVO_00843773	QORVO_00843777	Qorvo Termination Process dated 2023
85	QORVO_00843602	QORVO_00843605	Qorvo Human Resources Internal Process Employee Resignations dated 10/26/2022
86	QORVO_00888151	QORVO_00888157	Qorvo Human Resources and Cyber Security End User Data Access Reviews (Draft) dated 2021

DX	Beginning Bates	End Bates	Description
87	QORVO_00854273	QORVO_00854273	Digital Guardian Events for Investigation v2.0
88	QORVO_00854278	QORVO_00854278	Digital Guardian Open ARC Incidents
89	QORVO_00854274	QORVO_00854274	Data Access Review created by Lyndie Mitchell to review employee Clark Ragan dated 7/17/2023
90	QORVO_00854272	QORVO_00854272	Security Operations assigned to Jeremy Carrasquillo dated 8/3/2023
91	QORVO_00854277	QORVO_00854277	Spreadsheet of the configuration settings for the policies and rules in Digital Guardian authored by Jeremy Carrasquillo
92	QORVO_00843002	QORVO_00843035	Email from Olivia Marsh to Greg Warder dated 9/9/2021 re: IP Training - General Slides
93			Intentionally Omitted
94			Intentionally Omitted
95			Intentionally Omitted
96	QORVO_00843343	QORVO_00843343	Spreadsheet – “baw-Permissions matrix report.xlsx”
97	QORVO_00887805	QORVO_00887805	Preliminary QPF4656 Wi-Fi 6E Front End Module specifications sheet
98	QORVO_00887806	QORVO_00887806	Spreadsheet of classification history for a "Private" Excel
99	QORVO_00888143	QORVO_00888145	Document was categorized as "Critical" and the subcategory is “Controlled Unclassified Information (CUI)” then “Export Control,” designated for “No Foreign Dissemination.”
100	QORVO_00888146	QORVO_00888146	Bolden James classification history on a QPF4506B evaluation board test plan classified as “Private”

DX	Beginning Bates	End Bates	Description
101	QORVO_00866719	QORVO_00866747	Email from Todd Gillenwater to Eric Watkins dated 11/13/2018 re: RE: [External Email] RE: [External Email] [External Email] RE: [External Email] Call Follow-Up
102	QORVO_00866052	QORVO_00866053	Email from Eric Brentzel to Todd Gillenwater and James McEldowney re: RE: Urgent - Info Needed - Qorvo Infosec
103	QORVO_00844074	QORVO_00844081	Email from Bert Schumann to Gernot Fattinger and Robert Aigner dated 6/2/2015 re: FW: SA Insight: Akoustis Sets Out to Unsettle Established RF Filter Supply Base
104	QORVO_00844541	QORVO_00844542	Email from Gernot Fattinger to Robert Aigner dated 5/30/2016 re: Re: RFIC paper
105	QORVO_00844326	QORVO_00844342	Email from Eid Alsabbagh to Robert Aigner, Gernot Fattinger, and Otto Berger dated 9/23/2016 re: FW: 5G BPF spec for your review (new BAW filter supplier)
106	QORVO_00066840	QORVO_00066850	Email from Leah Giovan to Cees Links, Brian Balut, Tony Testa, and Gernot Fattinger dated 9/7/2017 re: Re: Akoustis Technologies Ships First 5.2GHz BAW RF Filter Prototypes to Multiple WiFi Customer Premise Equipment (CPE) and Semiconductor Companies - NADAW.com
107	QORVO_00844540	QORVO_00844539	Email from Robert Aigner to Gernot Fattinger dated 9/25/2017 re: FW: Akoustis 5.2 GHz evaluation board
108	QORVO_00314112	QORVO_00314122	Email from Leah Giovan to Roger Hall, Brian Balut, and Alex Zajac dated 12/12/2017 re: FW: Rev B & Rev C?
109	QORVO_00844633	QORVO_00844639	Email from Robert Aigner to Gernot Fattinger dated 5/18/2018 re: FW: CS ManTech Conference & ROCS Workshop Debrief - Friday, May 18, East Meeting Area, 10AM-Noon
110	AKTS_00041207	AKTS_00041208	Email from Dave Aichele to Gernot Fattinger dated 6/12/2018 re: Re: IMS Akoustis

DX	Beginning Bates	End Bates	Description
111	QORVO_00844538	QORVO_00844538	Email from Gernot Fattinger to Robert Aigner dated 6/5/2018 re: FW: IMS Akoustis
112	QORVO_00653753	QORVO_00653753	Email from Eid Alsabbagh to Wireless Connectivity Marketing, Wireless Connectivity MT, Alex Zajac, Gernot Fattinger, Mudar AlJoumayly, and Jeff Mink dated 8/1/2019 re: Fwd: Akoustis Announces Design Lock and Pre-Production Of 5.6 GHz WiFi BAW Filter Product
113	QORVO_0077016	QORVO_0077017	Email from Shawn Gibb to Gernot Fattinger dated 8/13/2019 re: RE: Great to meet you last week
114	QORVO_0062558	QORVO_0062634	Email from Shawn Gibb to Gernot Fattinger dated 8/12/2019 re: RE: Great to meet you last week
115	QORVO_00173941	QORVO_00173942	Email from Tony Testa to Jeff Mink, Gernot Fattinger, Alex Zajac, Cees Links, and Mudar AlJoumayly dated 9/6/2019 re: RE: Akoustis Datasheets (5.2G & 5.6G)
116	QORVO_00065107	QORVO_00065128	Email from John Anderton to Wayne Polonio dated 9/26/2019 re: FW: Akoustis
117	QORVO_00061033	QORVO_00061056	Email from Jeff Mink to FL IDP Acoustics Engineering and Shawn Gibb dated 10/23/2019 re: FW: Akoustis Datasheets (5.2G & 5.6G)
118	QORVO_00061515	QORVO_00061520	Email from Shawn Gibb to Gernot Fattinger dated 1/13/2020 re: FW: Akoustis Receives First 5G Network Infrastructure Filter Order for a Tier-1 Customer for 5G Small Cell Architecture - Akoustis Technologies, Inc. (AKTS)
119	QORVO_00031001	QORVO_00031003	Email from Gernot Fattinger to Bob Bruggeworth, Christina Lepiane, and Todd Gillenwater dated 7/14/2020 re: Re: AKTS
120	QORVO_00030994	QORVO_00031000	Email from Bob Bruggeworth to Tony Testa dated 7/14/2020 re: Re: AKTS
121	QORVO_00652846	QORVO_00652847	Email from Tony Testa to Cees Links dated 7/24/2020 re: FW: Resonant 6GHz

Dx	Beginning Bates	End Bates	Description
122	QORVO_00027672	QORVO_00027678	Email from David Schnauffer to Robert Aigner, Gernot Fattinger, Gareth Edwards, Alexis Mariani, Shawn Givv, Katie Caballero, Mudar AlJoumayly, and Tony Testa dated 10/16/2020 re: MWJ Acoustic Filter Market Analysis
123	QORVO_00063922	QORVO_00063923	Email from Issam Khayo to Cees Links, Cindy Warschauer, and Wireless Connectivity MT dated 3/1/2021 re: RE: Akoustis fab/labs BAW promo article
124	QORVO_00029952	QORVO_00029954	Email from Wilco Van Hoogstraeten to Gernot Fattinger dated 3/1/2021 re: RE: Akoustis fab/labs BAW promo article
125	QORVO_00029968	QORVO_00029970	Email from Bob Baeten to Tony Testa and John Anderton dated 3/31/2021 re: RE: Akoustis 6E filters prelim DS
126	QORVO_00849989	QORVO_00849990	Email from Paul Fego to Gernot Fattinger dated 8/14/2020 re: RE: Oppenheimer on AKTS
127	QORVO_00849634	QORVO_00849638	Email from Kevin Schoenrock to Gernot Fattinger dated 6/14/2023 re: RE: Akoustis Launches State-of-the-Art XBAW Foundry Services and AI-Enabled Engineering Design Software
128	QORVO_00849488	QORVO_00849512	Email from Gernot Fattinger to Mudar Al-Joumayly dated 8/14/2023 re: Fwd: Filter review meeting with Bob before end of Aug
129	QORVO_00850155	QORVO_00850170	Email from Scott Muir to Gernot Fattinger dated 8/22/2023 re: RE: AkST Competitive S Params
130	QORVO_00849921	QORVO_00849922	Email from Mudar Al-Joumayly to Gernot Fattinger dated 8/31/2023 (no subject)
131	QORVO_00664552	QORVO_00664553	Email from Gernot Fattinger to Bernard de Neeve, Alex Zajac, Bob Baeten, Connie MacKenzie, Dana Tunno, Eid Alsabbagh, Jeff Galipeau, Jeff Mink, Michael O'Neal, Otto Berger, Robert Aigner, and Rohan Houlden dated 1/18/2016 re: Re: 5 GHZ BAW Development (WiFi) - Meeting Summary - 01/18

DX	Beginning Bates	End Bates	Description
132	QORVO_00662434	QORVO_00662436	Email from Gernot Fattinger to Scott Knapp dated 1/20/2016 re: Re: 5 GHZ BAW Development (WiFi) - Meeting Summary - 01/18
133	QORVO_00847080	QORVO_00847112	Email from Eid Alsabbagh to Bernard de Neeve, Rohan Houlden, Tony Testa, Scott Knapp, Jeff Mink, Jeff Galipeau, Alex Zajac, Robert Qaigner, Otto Berger, and Gernot Fattinger dated 2/22/2016 re: RE: 5G GHZ Weekly meeting
134	QORVO_00687362	QORVO_00687449	Qorvo IDP Technology Roadmaps dated 3/10/2016
135	QORVO_00062938	QORVO_00062967	Qorvo Filter Technology Updated (undated)
136	QORVO_00653797	QORVO_00653800	Email from Cees Links to Gernot Fattinger and Tony Testa dated 5/10/2019 re: RE: 5 GHz BAW requirement
137	QORVO_00032666	QORVO_00032668	Email from Cees Links to Gernot Fattinger dated 9/17/2019 re: RE: Oppenheimer on AKTS
138	QORVO_00135198	QORVO_00135204	Email from Doug Bostrom to Tony Testa, Alex Zajac, and Wilco Van Hoogstraeten dated 12/16/2019 re: FW: 5.6 GHz filters update
139	QORVO_00627641	QORVO_00627644	Email from Roger Hall to Alex Zajac dated 12/21/2019 re: RE: Filter strategy: Follow up items
140	QORVO_00062820	QORVO_00062821	Email from Shawn Gibb to Gernot Fattinger dated 3/24/2020 re: RE: HPS - BAW GEN5W stack development
141	QORVO_00850043	QORVO_00850048	Email from Gernot Fattinger to Cees Links dated 2/19/2020 re: Re: 5.2 GHz BAW LTR Gate closed!
142	QORVO_00849929	QORVO_00849932	Email from Paul Fego to Gernot Fattinger dated 8/5/2020 re: FW: Input
143	QORVO_00804805	QORVO_00805002	Qorvo GSO WiFi Design Meeting dated 8/11/2021
144	QORVO_00000285	QORVO_00000432	USPTO Patent Number: 9735755 Issue date: 8/15/2017 titled "BAW Resonator Having Lateral Energy Confinement and Methods of Fabrication Thereof"

DX	Beginning Bates	End Bates	Description
145	AKTS_00462420	AKTS_00462425	BAW Resonator Design Considerations - An Overview (white paper) authored by Genrot G. Fattinger dated 2008
146	QORVO_00850876	QORVO_00850876	Email from Michael Boyd to Gernot Fattinger dated 4/5/2022 re: Action Item - One-Premise SharePoint Site Cleanup
147	QORVO_00850877	QORVO_00850877	Attachment to DTX-0146. Workbook of worksheets with data details of on-premise SharePoint sites dated 2022
148	QORVO_00850051	QORVO_00850051	Email from Gernot Fattinger to Michael Boyd dated 5/4/2022 re: Re: Insider Threat/Data Protection Audit Recommendations (Doug Bostrom, James Eastham, Gernot Fattinger)
149	QORVO_00850051	QORVO_00850051	Attachment to DTX-0148. Workbook of worksheets with data details of on-premise SharePoint sites dated 2022
150	QORVO_00849522	QORVO_00849524	Email from Angi Foreshaw to Josh Adams, James Eastham, Gernot Fattinger, and Doug Bostrom dated 6/25/2021 re: RE: SharePoint Content Shared w/ "Everyone" at Qorvo (Online & On Prem)
151	N/A	N/A	Spreadsheet attachment to DTX-0150, Email from Angi Foreshaw to Josh Adams, James Eastham, Gernot Fattinger, and Doug Bostrom dated 6/25/2021 re: RE: SharePoint Content Shared w/ "Everyone" at Qorvo (Online & On Prem)
152	QORVO_00844124	QORVO_00844125	Email from Gernot Fattinger to Alberto Canabal dated 8/19/2021 re: Re: Kamran Cheema joins Akoustis from QCOM RF360
153	QORVO_00844126	QORVO_00844126	Email from Gernot Fattinger to Ralph Rothmund, Mudar AlJoumayly, Doug Bostrom, and Vijay Balakrishna dated 5/3/2022 re: AKTS earnings call
154	QORVO_00844127	QORVO_00844128	Email from Gernot Fattinger to Doug Bostrom dated 5/5/2022 re: Re: AKTS earnings call
155	QORVO_00634749	QORVO_00634845	Qorvo Tech Day Session 1 dated 5/20/2020

DX	Beginning Bates	End Bates	Description
156	QORVO_00635359	QORVO_00635393	Qorvo Technology Roadmap for Newton 2025-2027 BAW draft (undated)
157	QORVO_00849595	QORVO_00849602	Email from Alfred Gimenez to Milad Koohi dated 6/13/2023 re: Re: Akoustis measured 10-GHz filter
158	QORVO_00062875	QORVO_00062925	Qorvo 5.25GHz Filter dated 1/25/2019
159	QORVO_00653761	QORVO_00653774	Email from Jeff Mink to Alex Zajac, Cees Links, Tony Testa, Gernot Fattinger, and Mudar AlJoumayly dated 9/6/2019 re: Akoustis Datasheets (5.2G & 5.6G)
160	QORVO_00652440	QORVO_00652442	Email from Bob Bruggeworth to Tony Testa dated 7/14/2020 re: Re: AKTS
161	QORVO_00665320	QORVO_00665321	Email from Brian Balut to Leah Giovan dated 12/4/2018 re: Re: Cisco building with 5GHz BAW next month
162	QORVO_00079503	QORVO_00079724	Qorvo Newton Tech Day 2022 dated 4/24/2022
163			Intentionally Omitted
164	GIOVAN 00001	GIOVAN 00008	Severance Agreement and Release of All Claims between Leah Giovan and Qorvo US, Inc. dated 5/4/2022
165	QORVO_00667064	QORVO_00667064	Email from Leah Giovan to Tony Testa dated 9/6/2017 re: Re: Akoustis Technologies Ships First 5.2GHz BAW Rf Filter Prototypes to Multiple WiFi Customer Premise Equipment (CPE) and Semiconductor Companies - NASDAQ.com
166	AKTS_00076955	AKTS_00076957	Email from Dave Aichele to Leah Giovan dated 10/11/2017 re: Re: Akoustis 5.2GHz evaluation board
167	QORVO_00314111	QORVO_00314111	Email from Leah Giovan to Todd Gillenwater, Brian Balut, Alex Zajac, Eric Creviston, and James Klein dated 11/28/2017 re: Akoustis 5GHz Filter Update
168	AKTS_00120745	AKTS_00120749	Email from Dave Aichele to Leah Giovan dated 12/8/2017 re: Re: Rev B & Rev C?
169	AKTS_00077474	AKTS_00077487	Email from Dave Aichele to Leah Giovan dated 12/11/2017 re: Re: Rev B & Rev C?

DX	Beginning Bates	End Bates	Description
170	AKTS_00077488	AKTS_00077491	Email from Dave Aichele to Leah Giovan dated 12/12/2017 re: Re: Rev B & Rev C?
171	AKTS_00214690	AKTS_00214694	Email from Dave Aichele to Leah Giovan dated 7/24/2017 re: NDA
172	AKTS_00076581	AKTS_00076588	Email from Dave Aichele to Leah Giovan dated 8/2/2017 re: Re: NDA
173	AKTS_00036258	AKTS_00036270	Email from Dave Aichele to Leah Giovan dated 8/22/2017 re: Re: NDA
174			Intentionally Omitted.
175			Intentionally Omitted.
176	AKTS_00078840	AKTS_00078843	Email from Dave Aichele to Leah Giovan dated 4/4/2018 re: Re: Akts AKF-1252 Summary
177	QORVO_00066562	QORVO_00066563	Email from Leah Giovan to Tony Testa dated 10/30/2018 re: FW: Call Follow-Up
178	QORVO_00066565	QORVO_00066567	Email from Leah Giovan to Jeffrey Shealy dated 11/1/2018 re: Re: [External Email] Call Follow-Up
179	QORVO_00066564	QORVO_00066564	Email from Leah Giovan to Cees Links dated 11/4/2018 re: Re: [External Email] Call Follow-Up
180	QORVO_00313877	QORVO_00313877	Email from Leah Given to Eric Creviston, Mark Murphy, Jim Stilson, Steve Grant, Bob Bruggeworth, Todd Gillenwater, and Hans Schwarz dated 1/30/2018 re: Akoustis
181	AKTS_00081038	AKTS_00081047	Email from Dave Aichele to Leah Giovan dated 8/30/2018 re: Re: [External Email] RE: Akts AKF-1252 Summary
182	AKTS_00082516	AKTS_00082517	Email from Jeffrey Shealy to Leah Giovan dated 10/2/2018 re: Re: [External Email] Akoustis_WCON Spec.pptx
183	AKTS_00128749	AKTS_00128754	Email from Jeffrey Shealy to Leah Giovan dated 10/4/2018 re: Re: [External Email] FW: Akoustis 5GHz filtering

DX	Beginning Bates	End Bates	Description
184	QORVO_00066578	QORVO_00066582	Email from James Klein to Paul Fego dated 10/31/2018 re: Fwd: C-V2X/DSRC coexistence Filter
185	AKTS_00084472	AKTS_00084480	Email from Jeffrey Shealy to Leah Giovan dated 11/12/2018 re: Re: [External Email] Re: [External Email] RE: [External Email] RE: [External Email] RE: [External Email] Call Follow-Up
186	QORVO_00319128	QORVO_00319128	Email from Hans Schwarz to Leah Giovan dated 12/10/2018 re: Akoustis update
187	QORVO_00135143	QORVO_00135143	Email from Leah Giovan to Tony Testa dated 2/2/2019 re: Re: Akoustis
188	AKTS_00085560	AKTS_00085588	Email from Jeffrey Shealy to Todd Gillenwater dated 2/7/2019 re: Re: Call Follow-Up
189	QORVO_00135144	QORVO_00135171	Email from Tony Testa to Leah Giovan dated 2/7/2019 re: RE: Call Follow-Up
190	AKTS_00085830	AKTS_00085831	Email from Jeffrey Shealy to Leah Giovan dated 3/1/2019 re: Re: AKTD-QRVO mtg at MWC?
191	AKTS_00086133	AKTS_00086137	Email from Jeffrey Shealy to Todd Gillenwater dated 3/22/2019 re: Re: Stack PDK's
192	QORVO_00653756	QORVO_00653759	Email from Cees Links to Gorden Cook, Tony Testa, James Klein, Brian Balut, and Wilco Van Hoogstraeten dated 4/4/2019 re: RE: AKTS press release
193	QORVO_00065502	QORVO_00065503	Email from Leah Giovan to Tony Testa dated 5/9/2019 re: FW: Oppenheimer note on AKTS mentions QRVO twice
194	QORVO_00121249	QORVO_00121278	Email from Alex Zajac to Tony Testa, Eid Alsabbagh, Wayne Polonio, and John Anderton dated 11/26/2019 re: FW: RFW34 HPS PL Weekly Report
195	QORVO_00625270	QORVO_00626279	Email from Jeff Mink to Matt Chun and Shawn Gibb dated 12/5/2019 re: FW: RFW34 HPS PL Weekly Report

DX	Beginning Bates	End Bates	Description
196	QORVO_00625905	QORVO_00625910	Email from Tuan Nguyen to Shawn Gibb dated 1/13/2020 re: FW: Akoustis Receives First 5G Network Infrastructure Filter Order for a Tier-1 Customer for 5G Small Cell Architecture - Akoustis Technologies, Inc. (AKTS)
197	QORVO_00627394	QORVO_00627394	Email from Tuan Nguyen to Shawn Gibb dated 1/27/2020 re: RE: Collins Buying Akoustis Filters
198	QORVO_00031052	QORVO_00031056	Email from Leah Giovan to James Klein, Cees Links, Doug Bostrom, and Tony Testa dated 5/5/2020 re: FW: More happy problems
199	QORVO_00308913	QORVO_00308916	Email from Leah Giovan to Hans Schwarz dated 9/10/2019 re: RE: Murata
200	QORVO_00172343	QORVO_00172343	Email from Tony Testa to Wayne Polonio dated 11/2/2016 re: Re: RH Landing Spot
201	QORVO_00137868	QORVO_00137869	Email from Alex Zajac to Wilco Van Hoogstraeten, Tony Testa, Cees Links, and Bob Baeten dated 3/6/2017 re: RE: AKTS: Two Customers Signed and More in the Works
202	QORVO_00312050	QORVO_00312052	Email from Cees Links to Hans Schwarz, Tony Testa, and David Lacinski dated 8/30/2018 re: RE: OPCO: AKTS (OUTPERFORM)- Initiating Coverage with Outperform and \$10 PT
203	QORVO_00065438	QORVO_00065501	Email from Leah Giovan to James Klein, Roger Hall, Doug Bostrom, and Tony Testa dated 3/20/2019 re: RE: Akoustis
204	QORVO_00135177	QORVO_00135178	Email from Eid Alsabbagh to Tony Testa dated 4/3/2019 re: RE: First 80,000 Units of Akoustis' 5.2 GHz WiFi Filter Products Shipped to Customer -
205	QORVO_00135183	QORVO_00135184	Email from Eid Alsabbagh to Cees Links dated 8/16/2019 re: RE: Akoustis Announces Design Lock and Pre-Production Of 5.6 GHz WiFi BAW Filter Product
206	QORVO_00032601	QORVO_00032603	Email from Bob Bruggeworth to Tony Testa dated 9/19/2019 re: RE: Akoustis

DX	Beginning Bates	End Bates	Description
207	QORVO_00032604	QORVO_00032606	Email from Bob Bruggeworth to Tony Testa dated 9/19/2019 re: re: Akoustis
208	QORVO_00082852	QORVO_00082853	Email from Tony Testa to Roger Hall dated 12/5/2019 re: Re: Akoustis Receives First 5G Network Infrastructure Filter Order for a Tier-1 Customer for 5G Small Cell Architecture - Akoustis Technologies, Inc. (AKTS)
209	QORVO_00065020	QORVO_00065026	Email from Alex Zajac to John Anderton, Eid Alsabbagh, Jeff Mink, and Wayne Polonio dated 1/23/2020 re: RE: QPQ1094MRD and PRD
210	QORVO_00065064	QORVO_00065087	Email from John Anderton to Eid Alsabbah, Alex Zajac, Jeff Mink, and Wayne Polonio dated 1/23/2020 re: RE: QPQ1904 MRD and PRD
211	QORVO_00122471	QORVO_00122476	Email from John Anderton to Cees Links, Doug Bostrom, Alex Zajac, and Tony Testa dated 9/1/2020 re: RE: Akoustis Announces Industry's first 6.5 GHZ filter
212	QORVO_00064861	QORVO_00064884	Email from Alex Zajac to Bob Baeten, Tony Testa, Brad Nelson, and Tuan Nguyen dated 9/9/2020 re: RE: QTEQ invited paper - IDP Competitive Analysis
213	QORVO_00122231	QORVO_00122232	Email from John Anderton to Tony Testa and Bob Baeten dated 3/26/2021 re: RE: Akoustis 6E filters prelim DS
214	QORVO_00063195	QORVO_00063205	Email from Alex Zajac to John Anderton and Cees Links dated 4/12/2021 re: RE: 6E Update
215	QORVO_00665132	QORVO_00665133	Email from Ahmad Abdelmajid to Tony Testa dated 9/27/2016 re: RE: 5G BPF spec for your review (new BAW filter supplier)
216	QORVO_00066560	QORVO_00066563	Email from Tony Testa to Cees Links and Wilco Van Hoogstraeten dated 11/3/2018 re: FW: Call Follow-Up
217	QORVO_00066575	QORVO_00066577	Email from Leah Giovan to Cees Links dated 11/5/2018 re: RE: Akoustis

DX	Beginning Bates	End Bates	Description
218	QORVO_00663217	QORVO_00663221	Email from Bob Baeten to Cees Links, Tony Testa, and Alex Zajac dated 12/22/2016 re: RE: IDP Project List for FY18 AOP input
219	QORVO_00092434	QORVO_00092436	Email from Tony Testa to Cees Links dated 5/9/2019 re: RE: 5GHz BAW requirement
220	QORVO_00092437	QORVO_00092438	Email from Tony Testa to Alex Zajac dated 5/9/2019 re: RE: 5GHz BAW requirement
221	QORVO_00092439	QORVO_00092440	Email from Tony Testa to Wilco Van Hoogstraeten, Cees Links, Alex Zajac, Gernot Fattinger, Brian Balut, Doug Bostrom, Robert Aigner, and Ganadhar Burra dated 5/9/2019 re: RE: 5GHz BAW requirement
222	QORVO_00135190	QORVO_00135192	Email from Eid Alsabbagh to Tony Testa dated 5/9/2019 re: RE: 5GHz BAW requirement
223	QORVO_00092441	QORVO_00092443	Email from Tony Testa to Cees Links dated 5/11/2019 re: RE: 5GHz BAW requirement
224	QORVO_00092444	QORVO_00092446	Email from Tony Testa to Cees Links dated 5/11/2019 re: RE: 5GHz BAW requirement
225	QORVO_00092447	QORVO_00092454	Email from Tony Testa to Alex Zajac, Brad Nelson, Bob Baeten, Kevin Bogacki, and Jeff Mink dated 5/14/2019 re: RE: 5GHz BAW requirement
226	QORVO_00122524	QORVO_00122531	Email From John Anderton to Cees Links, Tony Testa, Wayne Polonio, and Ed Alsabbagh dated 12/16/2019 re: RE: 15 Dec 2019 Program Management Wekkly for WCON
227	QORVO_00134620	QORVO_00134623	Email from Tony Testa to Wayne Polonio, Doug Bostrom, and Tommy Tran dated 12/19/2019 re: RE: Amazon - > 5.2 and 5.6 GHz filter samples need
228	QORVO_00122515	QORVO_00122520	Email from John Anderton to Wayne Polonio and Alex Zajac dated 12/23/2019 re: RE: BAW 5.6GHz news
229	QORVO_00166493	QORVO_00166499	Email from Tony Testa to Jeff Mink, Alex Zajac, John Anderton and Wayne Polonio dated 1/23/2020 re: RE: QPQ1904 MRD and PRD

DX	Beginning Bates	End Bates	Description
230	QORVO_00029242	QORVO_00029251	Qorvo 5GHz and FG Status Update BAW R&D dated 1/28/2020
231	QORVO_00082146	QORVO_00082153	Email from Tony Testa to Cees Links and Igor Lalicevic dated 3/21/2020 re: RE: APPROVAL NEEDED - Sagemcom - BT HH7 / Bell HH4k / Orange Poland and Spain
232	QORVO_00095807	QORVO_00095811	Anthony (Tony) Testa Jr.'s Voyage: Overall Section Year-End Check-In & Summary Information dated 3/21/2020
233			Intentionally Omitted.
234	QORVO_00063827	QORVO_00063829	Email from Tony Testa to Cees Links, Bart Dehouwer, Bettina Meles, Bob Baeten, Jaidev Sharma, John Anderton, Tim Allemeersch, and Wilco Van Hoogstraeten dated 4/20/2021 re: RE: APAC FAE Activities Update - CW16, 2021
235	QORVO_00557028	QORVO_00557029	Email from Kevin Schoenrock to Wayne Polonio, Tony Testa, and Bob Baeten dated 2/26/2023 re: RE: AKTS filter performance feedback
236	QORVO_00556901	QORVO_00556901	Qorvo analysis of its products and its customers' needs
237	QORVO_00654057	QORVO_00654062	Email from Andrew Ketterson to Qorvo employees dated 9/10/2019 re: IDP Technology Portfolio Review September '19 - Actions and minutes
238	QORVO_00627595	QORVO_00627600	Email from Andrew Ketterson to Qorvo employees dated 3/16/2020 re: Minutes and Actions: IDP Technology Portfolio Review - MAR
239	QORVO_00627710	QORVO_00627720	Email from Andrew Ketterson to Qorvo employees dated 5/6/2020 re: Minutes and Actions: IDP Technology Portfolio Review - MAY
240	QORVO_00627543	QORVO_00627554	Email from Andrew Ketterson to Qorvo employees dated 1/18/2021 re: IDP Technology Portfolio Review - JAN Minutes

DX	Beginning Bates	End Bates	Description
241	QORVO_00627669	QORVO_00627682	Email from Andrew Ketterson to Qorvo employees dated 3/20/2021 re: IDP Technology Portfolio Review - MAR Minutes and Actions
242	QORVO_00688805	QORVO_00688807	Email from Maurice (Cheng) Huang to Qorvo Employees dated 3/10/2020 re: RE: BAW 5.2, 5.6, 5.9 Samples
243	QORVO_00653836	QORVO_00653844	Email from Eid Alsabbagh to Cees Links dated 12/17/2019 re: RE: 15 Dec 2019 Program Management Weekly for WCON
244	QORVO_00063822	QORVO_00063826	Email from Bob Baeten to Cees Links, Jeffrey Yang, and Jaidev Sharma dated 2/9/2021 re: RE: APAC FAE Activities update - CW06, 2021
245			Intentionally Omitted.
246	QORVO_00694742	QORVO_00694742	Email from Bob Baeten to Cheng-Hui Lin, Tony Testa, Ahmad Abdelmajid and Kevin Bogacki dated 7/14/2016 re: RE: SKY new 5G 3x5 mm2 HP FEM - SKY85743-11
247	QORVO_00651271	QORVO_00651292	Email from Bob Baeten to Kevin Bogacki, Tony Testa, Eid Alsabbagh, and Ahmad Abdelmajid dated 2/19/2015 re: FW: Microsemi Product Roadmap-Feb 2014
248	QORVO_00808527	QORVO_00808529	Email from Kevin Bogacki to Cody Hale dated 7/8/2016 re: FW: Comcast/Quantenna
249	QORVO_00136743	QORVO_00136745	Email from Wayne Polonio to Tony Testa dated 4/27/2018 re: Fw: 5GHz BAW
250	QORVO_00093438	QORVO_00093438	Email from Jill Brunato to Wayne Polonio, Tommy Tran, Jeff Mink, and Alex Zajac dated 12/6/2018 re: Akoustis 5G BAW specs
251	QORVO_00064000	QORVO_00064003	Email from Igor Lalicevic to Jeff Lin, Corrine Chen, Ivan Wei, Wireless Connectivity MT, Wireless Connectivity. PLCm, and Wcon.GWMarketing dated 3/3/2020 re: Minutes & Actions: Sagemcom France, March 2nd, 2020

DX	Beginning Bates	End Bates	Description
252	QORVO_00649999	QORVO_00665067	Email from Kevin Bogacki to Qorvo Employees dated 3/4/2016 re: [Competitive Analysis] - SKY75410-11 5GHz 4x4mm PA
253	QORVO_00637461	QORVO_00637465	Email from Eid Alsabbagh to Wayne Polonio, Tony Testa, and Ken Arizpe dated 3/18/2015 re: Fwd: Emailing: Qualcomm Small Cell NDA Roadmap (March 2015).pdf
254	QORVO_00643216	QORVO_00643290	Email from Eid Alsabbagh to Tony Testa dated 8/14/2015 re: RE: Product Initiation
255	QORVO_00063831	QORVO_00063834	Email from Brock West to Tony Testa dated 3/16/2021 re: RE: Design Losses
256	QORVO_00063153	QORVO_00063155	Email from Michelle Rauh to Tony Testa dated 2/7/2022 re: RE: filter competitor data- Scrap report I will send new one
257	QORVO_00063162	QORVO_00063173	Email from Jeremy Foland to Tony Testa, Jeff Lin, and Wcon.GWMarketing dated 2/11/2022 re: RE: Filter competitor analysis
258	QORVO_00133852	QORVO_00133854	Email from Tony Testa to Bob Bruggeworth dated 5/5/2020 re: RE: Akoustis Announces First Tri-Bank WiFi CPE Design Win and Filter Orders with Tier-1 OEM - Akoustis Technologies, Inc. (A...
259			Intentionally Omitted
260	QORVO_00811567	QORVO_00811733	Email from Kevin Bogacki to Andy Hazelwood dated 1/28/2015 re: FW: SKY85717 test report
261	QORVO_00651191	QORVO_00651263	Email from Bob Baeten to Rohan Houlden and Tony Testa dated 7/1/2015 re: FW: SKYS roadmap
262	QORVO_00690086	QORVO_00690113	Qorvo FAE Intelligence Report - WCON dated Q4, 2018
263	QORVO_00752044	QORVO_00752082	Qorvo FAE Intelligence Report - WCON dated Q1, 2019
264	AKTS_00494322	AKTS_00494350	Qorvo FAE Intelligence Report - WCON dated Q2, 2020
265	AKTS_00157835	AKTS_00157868	Qorvo FAE Intelligence Report - WCON dated Q1, 2021

DX	Beginning Bates	End Bates	Description
266	QORVO_00838069	QORVO_00838090	Qorvo FAE Intelligence Report - WCON dated Q3, 2021
267	QORVO_00558516	QORVO_00558569	Email from Kevin Schoenrock to David Gonzales - Oregon, Kevin Gallagher, and David Drummond dated 4/16/2021 re: WiFi ppt and xls attaching Strat Team Market Study_WiFi rev04.pptx and WiFi Market Study rev04.16.xlsx
268	QORVO_00308664	QORVO_00308664	Email from Brandi Frye to Bob Bruggeworth, Brandi Frye, Eric Creviston, Gary Grant, Hans Schwarz, James Klein, Jamie Flinchbaugh, Mark Murphy, Paul Fego, and Ralph Knupp dated 10/3/2019 re: Exec Staff meeting notes 10/3/19
269	QORVO_00311156	QORVO_00311158	Email from Todd Gillenwater to Hans Schwarz dated 1/30/2019 re: Re: OPCO: AKTS (OUTPERFORM) - AKTS F2Q Earnings Preview
270	QORVO_00163190	QORVO_00163213	Email from Brock West to Bob Bruggeworth dated 10/23/2019 re: Earnings prep
271	QORVO_00700350	QORVO_00700352	Email from Wayne Polonio to Eid Alsabbagh dated 1/16/2020 re: RE: 5GHz BAW review collateral update
272	QORVO_00865113	QORVO_00865113	Email from Craig Callahan to Todd Gillenwater dated 8/27/2020 re: RE: Akoustis
273	QORVO_00174377	QORVO_00174377	Email from Wayne Polonio to Jeremy Foland, Tony Testa, Jeff Lin, Igor Lalicevic, and Kathy DaSilva dated 5/11/2022 (no subject)
274	QORVO_00843772	QORVO_00843772	Spreadsheet of an analysis by Qorvo of its opportunities and why they lost the bids.
275	QORVO_00849387	QORVO_00849387	Email from Otto Berger to Robert Aigner and Gernot Fattinger dated 4/13/2015 re: FW: 5GHz BAW Support
276	QORVO_00845710	QORVO_00845711	Email from Robert Aigner to Otto Berger dated 2/16/2016 re: RE: 5GHz WiFi BAW Stack - Need help
277	QORVO_00849176	QORVO_00849178	Email from Rohan Houlden to Robert Aigner dated 2/29/2016 re: 5 GHz BAW filters

DX	Beginning Bates	End Bates	Description
278	QORVO_00121590	QORVO_00121645	Email from Tony Testa to Qorvo Employees dated 9/17/2018 re: RE: Week of 17 Sep attaching WCON Strategic Planning MTG #3 FY Process dated 9/16/2018
279	QORVO_00172422	QORVO_00174242	Email from Wayne Polonio to Scott Muir dated 1/23/2020 re: RE: QPPQ1904 MRD and PRD
280	QORVO_00888703	QORVO_00888709	Email from Ramona L Mueller to Wayne Polonio and Bernard de Neeve dated 11/23/2020 re: RTS for Review: QPQ1903 & QPQ1994
281	QORVO_00063185	QORVO_00063187	Email from Tony Testa to Cees Links dated 2/1/2021 re: RE: Update on the Evaluation of Resonant's XBAR Filter Technology
282	QORVO_00598852	QORVO_00598854	Email from James Wilson to Kevin Schoenrock dated 8/21/2020 re: RE: Strategy Ideas
283	QORVO_00135651	QORVO_00135651	Email from Bob Baeten to Cees Links and Tony Testa dated 4/30/2019 re: FW: Organization Announcement
284	QORVO_00172387	QORVO_00172387	Email from Wayne Polonio to Eid Alsabbagh, Cees Links, and Tony Testa dated 4/24/2020 re: RE: Eero/Akoustis
285	QORVO_00121895	QORVO_00121898	Email from John Anderton to Colin Leong dated 11/9/2021 re: RE: WCON Weekly PLM Meeting
286	QORVO_00889260	QORVO_00889260	Native File - Report: SPR22656 - BAW Filter Comparison 2022 Reverse Coating dated August 2022
287	QORVO_00840845	QORVO_00840845	Native File - Qorvo CSG Latest Estimate FY 2023-11 (FEB) Billings (Forecasted)
288	QORVO_00842023	QORVO_00842023	Native File - Qorvo Inc and Subsidiaries Consolidated Statements of Income
289	QORVO_00843341	QORVO_00843341	Native File - Qorvo Latest Estimate FY 2024-04 (JUL) Billings (Forecasted)

DX	Beginning Bates	End Bates	Description
290	QORVO_00890496	QORVO_00890497	Email from Kevin Schoenrock to Tony Testa dated 11/5/2023 (no subject) attaching Print Mixed, but Guide Well Below on Multi-Quarter Inventory Correction; Cutting PT from \$9.25 to \$1.75, Rating from 8uy to Neutral.pdf Unfiltered Growth_ Initiating Coverage of RF Filter Specialist AKIS at Buy, \$9.25 PT.pdf, CPE WIFI TAM_TT_Nov23 _revised.xlsx
291	N/A	N/A	U.S. Patent No. US 9,537,465 B1 dated 1/3/2017 titled Acoustic Resonator Device with Single Crystal Piezo Material and Capacitor on a Bulk Substrate
292	N/A	N/A	U.S. Patent No. US 9,571,061 B2 dated 2/14/2017 titled Integrated Circuit Configured with Two or More Single Crystal Acoustic Resonator Devices
293	N/A	N/A	U.S. Patent No. US 9,673,384 B2 dated 6/6/2017 titled Resonance Circuit with a Single Crystal Capacitor Dielectric Material
294	N/A	N/A	U.S. Patent No. US 9,716,581 B2 dated 7/25/2017 titled Mobile Communication Device Configured with a Single Crystal Piezo Resonator Structure
295	N/A	N/A	U.S. Patent No. US 9,805,966 B2 dated 10/31/2017 titled Wafer Scale Packaging
296	QORVO_00612453	QORVO_00612460	Akoustis Expands Commercial Product Portfolio with New 3.8 GHz BAW RF Filter article dated 4/9/2018
297			Intentionally Omitted
298	AKTS_00458291	AKTS_00458297	Akoustis Introduces New C-V2X Filter Solution for the Automotive Market article dated 4/11/23
299	AKTS_00081041	AKTS_00081047	Akoustis' 5.2 GHz WiFi Coexistence BAW Filter - Preliminary AKF-1252 dated 8/23/2018
300			Intentionally Omitted

DX	Beginning Bates	End Bates	Description
301	AKTS_00082942	AKTS_00082945	Email from Jeffrey Shealy to Hans Schwarz dated 10/18/2018 re: Re: [External Email] RE: [External Email] RE: [External Email] RE: [External Email] RE: Akoustis Strategic Inquiry
302	AKTS_00084310	AKTS_00084311	Email from Jeffrey Shealy to Leah Giovan dated 10/31/2018 re: Re: [External Email] Call Follow-Up
303	AKTS_00084421	AKTS_00084424	Email from Jeffrey Shealy to Leah Giovan dated 11/9/2018 re: Re: [External Email] RE: [External Email] Call Follow-Up
304	AKTS_00084425	AKTS_00084429	Email from Jeffrey Shealy to Leah Giovan dated 11/11/2018 re: Re: [External Email] RE: [External Email] RE: [External Email] Call Follow-Up
305			Intentionally Omitted
306	AKTS_00085383	AKTS_00085395	Email from Jeffrey Shealy to Leah Giovan dated 1/28/2019 re: Re: Call Follow-Up
307	AKTS_00085429	AKTS_00085456	Email from Jeffrey Shealy to Leah Giovan dated 1/29/2019 re: Re: Call Follow-Up
308			Intentionally Omitted
309	AKTS_00122854	AKTS_00122854	Email from Dave Aichele to Ichong@qkuantenna.com dated 3/20/2018 re: Akoustis Follow up
310			Intentionally Omitted
311	AKTS_00122856	AKTS_00122870	Akoustis WiFi BAW Filter Update dated March '18
312	AKTS_00128617	AKTS_00128617	Email from Dave Aichele to dongjoo.kim@samsung.com dated 10/1/2018 re: Akoustis Invoice & DDR
313	AKTS_00128618	AKTS_00128618	Akoustis Invoice # 0000118-IN to Samsung Electronics Co., Ltd. for one Detailed Design Review Milestone for \$25,000 dated 9/10/2018

DX	Beginning Bates	End Bates	Description
314	AKTS_00128619	AKTS_00128636	Akoustis Detailed Design Review - Phase 1: AKF-1435: 3.5 GHz discrete filler for 5G dated 9/20/2018
315	AKTS_00128719	AKTS_00128720	Email from Jeffrey Shealy to Hans Schwarz dated 10/2/2018 re: Re: Akoustis Strategic Inquiry
316	AKTS_00128721	AKTS_00128723	Draft term sheet between Akoustis Technologies, Inc. and Qorvo, Inc. concerning an equity investment and other potential business opportunities
317	AKTS_00128724	AKTS_00128724	Foundry Customer - RF Filter Design Flow (Customer designed XBAW RF filter) (undated)
318	AKTS_00137448	AKTS_00137448	Email from Annia Shen to Annia Shen dated 4/19/2019 re: AKF-1256 MSET073 Design Review_rev1.pptx
319	AKTS_00137449	AKTS_00137470	Akoustis MSET073 Design Review dated 4/19/2019
320	AKTS_00240442	AKTS_00240456	Email from Rohan Houlden to Annia Shen, Dave Aichele, and Ali Bawangaonwala dated 11/19/2018 re: 1256 10 EVB's ship next wk
321	AKTS_00243294	AKTS_00243303	Email from Ali Bawangaonwala to Rama Vetury, Joel Morgan, Rohan Houlden, and Dave Aichele dated 12/28/2018 re: 12/27: Weekly Product Meeting Minutes AKF-1256
322	AKTS_00270430	AKTS_00270458	Email from Jeffrey Shealy to Drew Wright dated 10/20/2019 re: FW: Call Follow-Up
323	AKTS_00270459	AKTS_00270470	Akoustis AKF-1252 Data Summary As Measured by Qorvo, Greensboro, NC dated 2/5/2019
324	AKTS_00311765	AKTS_00311765	Email from Dave Hodge to Emad Mehdizadeh dated 12/1/2020 (no subject)
325	AKTS_00339702	AKTS_00339728	Qorvo FAE Intelligence Report- WCON dated Q2, 2021
326	AKTS_00426416	AKTS_00426420	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between David M. Aichele and Akoustis, Inc. dated 5/6/2015

DX	Beginning Bates	End Bates	Description
327	AKTS_00426448	AKTS_00426452	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Akoustis, Inc. and Robert Dry dated 10/16/2019
328	AKTS_00426466	AKTS_00426470	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Todd R. Bender and Akoustis, Inc. dated 11/27/2017
329	AKTS_00426532	AKTS_00426536	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Akoustis, Inc. and Jeffrey Shealy dated 11/10/2020
330	AKTS_00426653	AKTS_00426657	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Wendy Wright and Akoustis, Inc. dated 3/11/2019
331	AKTS_00426668	AKTS_00426670	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Michael David Hodge and Akoustis, Inc. dated 7/13/2015
332	AKTS_00426683	AKTS_00426687	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Rohan W. Houlden and Akoustis, Inc. dated 9/19/2016
333	AKTS_00426699	AKTS_00426703	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Joonbum Kwon and Akoustis, Inc. dated 6/26/2017
334	AKTS_00426801	AKTS_00426805	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Ya Shen and Akoustis, Inc. dated 2/27/2017
335	AKTS_00426815	AKTS_00426819	Employee Confidentiality, Proprietary Information and Patent and Invention Assignment Agreement between Mary Winters and Akoustis, Inc. dated 6/26/2017
336	AKTS_00427862	AKTS_00427890	Akoustis presentation on design and performance of its AKS-103 product, which has a circular wafer

DX	Beginning Bates	End Bates	Description
337	AKTS_00428005	AKTS_00428010	Akoustis Resonator Failure Analysis dated 10/7/2014
338	AKTS_00439551	AKTS_00439555	Email from Dave Aichele to mjeong.ko@samsung.com dated 11/21/2016 re: Re: [B/L Vo. V0238422598] Standard NDA with Akoustis and Samsung Electromechanics
339	AKTS_00442266	AKTS_00442269	Akoustis, Inc.'s Purchase Order number 1000212 dated 11/26/2014 to EpiGaN nv in Belgium
340	AKTS_00442270	AKTS_00442287	Akoustis COMSOL Epi Simulation Update dated 12/23/2015
341	AKTS_00442288	AKTS_00442288	S-parameter performance plot for Akoustis COMSOL Epi Simulation Update dated 12/23/2015
342	AKTS_00442289	AKTS_00442289	S-parameter performance plot for Akoustis COMSOL Epi Simulation Update dated 12/23/2016
343	AKTS_00442290	AKTS_00442290	S-parameter performance plot for Akoustis COMSOL Epi Simulation Update dated 12/23/2017
344			Intentionally Omitted
345	AKTS_00442296	AKTS_00442301	Akoustis 13 wafers arrived (undated)
346			Intentionally Omitted
347	AKTS_00442307	AKTS_00442371	Electronic Acknowledgment Receipt for Invention Application for Membrane Substrate Structure for Single Crystal Acoustic Resonator Device dated 8/26/2014
348	AKTS_00442372	AKTS_00442373	Flat Capacitance, Conductance vs Voltage and 22pf capacitor in parallel with 1MOhm resistor
349	AKTS_00442374	AKTS_00442378	Electron response August 2015 For discussion purposes only
350	AKTS_00442379	AKTS_00442386	Akoustis Epitaxial Characterization - Single Crystal Acoustic Resonator (SCAR) dated 8/6/2014

DX	Beginning Bates	End Bates	Description
351	AKTS_00442387	AKTS_00442390	Akoustis, Inc.'s III-Nitride epi structures Request for Quote dated 6/9/2014
352	AKTS_00442391	AKTS_00442393	EpiGaN Quotation for HEMT Epilayers dated 11/7/14
353	AKTS_00442394	AKTS_00442394	Performance analysis of GaN
354	AKTS_00442395	AKTS_00442395	Spreadsheet of simulated performance metrics
355	AKTS_00442414	AKTS_00442414	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015
356	AKTS_00442415	AKTS_00442415	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2016
357	AKTS_00442416	AKTS_00442416	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2017
358	AKTS_00442417	AKTS_00442417	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2018
359	AKTS_00442418	AKTS_00442423	Akoustis, Inc.'s GaN epi structures Request for Quote dated 6/9/2014
360	AKTS_00442424	AKTS_00442425	Akoustis' letter to "To Whom it May Concern" regarding the contents of materials from EpiGaN in Belgium shipped via Fexex
361	AKTS_00442426	AKTS_00442426	Department of Homeland Security U.S. Customs and Border Protection Importer ID Input Record dated 7/16/2014
362	AKTS_00442445	AKTS_00442445	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015
363	AKTS_00442446	AKTS_00442446	Graph of admittance simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015
364	AKTS_00442447	AKTS_00442447	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015

DX	Beginning Bates	End Bates	Description
365	AKTS_00442448	AKTS_00442448	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015
366	AKTS_00442449	AKTS_00442449	Graph of s-parameter simulation results from Akoustis COMSOL Epi Simulation Update dated December 23, 2015
367	AKTS_00452030	AKTS_00452055	U.S. Patent Feldman et al. Patent No.: US 10,217,930 B1 dated 2/26/2019 titled "Method of Manufacture for Single Crystal Acoustic Resonator Devices Using Micro-Vias"
368	AKTS_00452165	AKTS_00452256	U.S. Patent Shealy et al. Patent No.: US 10,979, 022 B2 dated 4/13/2021 titled "5.2 GHZ Wi-Fi Acoustic Wave Resonator RF Filter Circuit"
369	AKTS_00457755	AKTS_00457846	U.S. Securities and Exchange Commission Form 8-K Current Report for Akoustis Technologies, Inc. dated 5/22/2015
370	AKTS_00458253	AKTS_00458257	Akoustis Board of Directors webpage (undated)
371	AKTS_00458258	AKTS_00458263	Akoustis Press Release of "Akoustis Announces BAW RF Filter Design Win and Commercial Production Order" dated 7/16/2018 webpage
372	AKTS_00458270	AKTS_00458274	Akoustis Press Release of "Akoustis Technologies to Attend 2019 Consumer Electronics Show" dated 12/31/2018 webpage
373	AKTS_00458275	AKTS_00458283	Akoustis Press Release of "Akoustis Acquires Majority Ownership Position of RFM Integrated Device, Inc., a Fabless Supplier of Acoustic Wave RF Resonators and Filters" dated 10/18/2021 webpage
374	AKTS_00458284	AKTS_00458290	Akoustis Announces Next Generation Wi-Fi 6E and Wi-Fi 7 Filter Solutions" dated 10/4/2022 webpage
375	AKTS_00458298	AKTS_099458306	Akoustis Press Release of "Akoustis Technologies Complete Strategic Acquisition of New York Wafer Manufacturing Facility and Operation" dated 6/27/2017 webpage

DX	Beginning Bates	End Bates	Description
376	AKTS_00458307	AKTS_00458307	Quarterly report pursuant to Section 13 of 15(d) dated 4/30/2015 regarding "Merger with Akoustis, Inc."
377	AKTS_00458346	AKTS_00458392	U.S. Securities and Exchange Commission Form S-1 for Danlax, Corp. dated 1/21/2014
378	AKTS_00851239	AKTS_00851261	Global Communication Semiconductor, LLC's PR-PFG-NEW FBAR Lot Follower Title: New FBAR Revision History
379	AKTS_01101416	AKTS_01101471	Akoustis Management Presentation dated October 2019
380	N/A	N/A	Article "Mitigating Spectrum Coexistence of V2X Systems with High-Performance Filters" authored by Ali Bawangaonwala (undated)
381	EXEC-TEK_00000027	EXEC-TEK_00000028	Email from Lora Shealy to Michael Ortiz dated 12/19/2016 re: FW: Revised Offer
382	EXEC-TEK_00000029	EXEC-TEK_00000032	Akoustis' Offer of employment to Dr. Ya "Annia" Shen
383	QORVO_00311765	QORVO_00311765	Email from Jeffrey Shealy to Todd Gillenwater dated 10/1/2018 re: RE: [External Email] Akoustis Strategic Inquiry
384			Intentionally Omitted
385	QORVO_00665311	QORVO_00665311	Email from Cees Links to Rohan Houlden dated 9/23/2018 re: Akoustis
386	N/A	N/A	Qorvo bandBoost Filters Enable Maximum Capacity and Throughput (undated)
387	QORVO_00000837	QORVO_00000840	News article about Akoustis completing tech. qualification for single crystal BAW RF Filter Products dated 7/9/2018
388	QORVO_00013168	QORVO_00013168	Email from R. Houlden to D. Aichele and R. Dry (Qorvo)
389	QORVO_00013171	QORVO_00013171	Email chain related to production and purchase orders
390	QORVO_00013323	QORVO_00013360	MicroWave Journal with articles about Qorvo and Akoustis

DX	Beginning Bates	End Bates	Description
391	QORVO_00013416	QORVO_00013422	Technical Datasheet for Akoustis Baw filter (A10335)
392	QORVO_00013726	QORVO_00013727	Email chain showing Qorvo docs sent to personal gmail by G. Moreno (Qorvo)
393	QORVO_00013728	QORVO_00013730	Qorvo presentation related to Murata product
394	QORVO_00013731	QORVO_00013733	Qorvo presentation related to Murata product
395	QORVO_00026833	QORVO_00026833	Email chain between Akoustis and David Benton (Qorvo)
396	QORVO_00027166	QORVO_00027168	Email exchange discussing Robert Dry moving to AKTS
397	QORVO_00027716	QORVO_00027722	Email exchange discussing Akoustis BAW filters
398	QORVO_00027783	QORVO_00027783	Internal Qorvo email about AKTS-1938
399	QORVO_00027790	QORVO_00027807	Qorvo BAW presentation
400	QORVO_00028210	QORVO_00028282	Email from Tuan Nguyen to HPS BTS CAMs dated 3/1/2021 re: Mobile Experts Transceiver report 02-2021, attaching Macro Base Station Transceivers 2021 from Mobile Experts
401	QORVO_00028290	QORVO_00028298	Internal Qorvo email about A10235 and A10149
402	QORVO_00028309	QORVO_00028312	Internal Qorvo email exchange about Akoustis' product development and Qorvo deciding to develop their own products to compete
403	QORVO_00028313	QORVO_00028316	Internal Qorvo email exchange about Akoustis vs Qorvo product development
404	QORVO_00028317	QORVO_00028321	Internal Qorvo email exchange about Akoustis vs Qorvo product development
405	QORVO_00028517	QORVO_00028517	Internal Qorvo email about Wendy Wright leaving Qorvo
406			Intentionally Omitted
407	QORVO_00029982	QORVO_00029996	Internal Qorvo email circulating Akoustis Datasheets

DX	Beginning Bates	End Bates	Description
408	QORVO_00030039	QORVO_00030041	Internal Qorvo email exchange discussing pricing
409	QORVO_00030105	QORVO_00030112	Market analytics report from Strategy Analytics
410	QORVO_00030129	QORVO_00030130	Letter from Qorvo to Akoustis
411	QORVO_00030140	QORVO_00030151	Internal Qorvo email exchange discussing acoustic filter market
412	QORVO_00030969	QORVO_00030971	Internal Qorvo emails about Akoustis BAW filters
413	QORVO_00031097	QORVO_00031097	Internal Qorvo email about developing 6.5GHz filters
414	QORVO_00032598	QORVO_00032600	Internal Qorvo email exchange about Akoustis product development
415	QORVO_00033506	QORVO_00033594	2016 IEEE RF Symposium
416	QORVO_00033647	QORVO_00033647	Qorvo email with attachment PPTX comparing Qorvo and Akoustis products
417	QORVO_00060801	QORVO_00060900	Qorvo Q3 2020 Quarterly Update
418	QORVO_00060987	QORVO_00060989	Internal Qorvo emails about Akoustis BAW filters
419	QORVO_00060990	QORVO_00060995	Internal Qorvo emails about filter strategy
420	QORVO_00061002	QORVO_00061008	Internal Qorvo emails discussing strategy against Akoustis
421	QORVO_00061036	QORVO_00061069	Qorvo presentation to compare Akoustis vs. Qorvo products
422	QORVO_00061108	QORVO_00061109	Internal Qorvo emails about hiring Shawn Gibb
423			Intentionally Omitted
424	QORVO_00061567	QORVO_00061570	Internal Qorvo emails about Akoustis 5G filters
425	QORVO_00061707	QORVO_00061743	Internal Qorvo emails about Qorvo filters, including comparing with Akoustis filters
426	QORVO_00061744	QORVO_00061744	Internal Qorvo emails about Akoustis filter pricing

DX	Beginning Bates	End Bates	Description
427	QORVO_00061745	QORVO_00061745	Internal Qorvo emails about Akoustis filter pricing
428	QORVO_00061879	QORVO_00061881	Internal Qorvo emails about Qorvo filters
429	QORVO_00061882	QORVO_00061916	Internal Qorvo emails about Qorvo filters
430	QORVO_00062161	QORVO_00062162	Internal Qorvo emails about Akoustis filters
431	QORVO_00062269	QORVO_00062274	Qorvo product presentation
432	QORVO_00062278	QORVO_00062283	Internal Qorvo emails about Akoustis filters
433	QORVO_00062527	QORVO_00062529	Internal Qorvo emails about losing business to Akoustis filters
434	QORVO_00062689	QORVO_00062696	Internal Qorvo emails about Akoustis IP strategy
435	QORVO_00062776	QORVO_00062777	Internal Qorvo emails about Akoustis filters
436	QORVO_00062815	QORVO_00062827	Internal Qorvo emails about Akoustis filters
437	QORVO_00062835	QORVO_00062839	Qorvo Q&A about the competitive BAW filter market
438	QORVO_00062926	QORVO_00062936	Akoustis PPT on AKF-1252P
439			Intentionally Omitted
440	QORVO_00062968	QORVO_00062970	Qorvo NDA with Akoustis
441			Intentionally Omitted
442	QORVO_00063150	QORVO_00063152	Internal Qorvo emails about Akoustis filters as competitors in the market
443			Intentionally Omitted
444	QORVO_00063156	QORVO_00063157	Internal Qorvo emails about Akoustis filters as competitors in the market

DX	Beginning Bates	End Bates	Description
445			Intentionally Omitted
446			Intentionally Omitted
447			Intentionally Omitted
448	QORVO_00063830	QORVO_00063830	Internal Qorvo email about hiring Akoustis engineers
449			Intentionally Omitted
450	QORVO_00063835	QORVO_00063839	Internal Qorvo email about losses to Akoustis
451	QORVO_00063956	QORVO_00063961	Qorvo email exchange about competing with Akoustis products
452	QORVO_00063962	QORVO_00063979	Qorvo email exchange about competing with Akoustis products
453	QORVO_00063980	QORVO_00063986	Qorvo email exchange about competing with Akoustis products
454	QORVO_00063987	QORVO_00063991	Qorvo email exchange about competing with Akoustis products
455	QORVO_00063992	QORVO_00063999	Qorvo email exchange about competing with Akoustis products
456			Intentionally Omitted
457			Intentionally Omitted
458			Intentionally Omitted
459			Intentionally Omitted
460			Intentionally Omitted
461			Intentionally Omitted
462			Intentionally Omitted

DX	Beginning Bates	End Bates	Description
463			Intentionally Omitted
464	QORVO_00065129	QORVO_00065131	Internal Qorvo emails about Akoustis process flow
465			Intentionally Omitted
466	QORVO_00065724	QORVO_00065732	Qorvo email exchange comparing products with Akoustis
467	QORVO_00066508	QORVO_00066511	Slides from Qorvo presentation
468			Intentionally Omitted
469	QORVO_00066598	QORVO_00066599	Internal Qorvo emails about Akoustis capabilities
470	QORVO_00066828	QORVO_00066829	Internal Qorvo emails about Akoustis competition
471			Intentionally Omitted
472	QORVO_00070764	QORVO_00070765	Qorvo email about Skyworks data
473	QORVO_00070776	QORVO_00070790	Skyworks presentation about products
474	QORVO_00070791	QORVO_00070811	Skyworks presentation about products
475	QORVO_00070812	QORVO_00070813	Internal Qorvo email about Skyworks
476			Intentionally Omitted
477			Intentionally Omitted
478			Intentionally Omitted
479			Intentionally Omitted
480	QORVO_00072324	QORVO_00072496	Yole report comparing BAW filters
481			Intentionally Omitted
482	QORVO_00072509	QORVO_00072516	Qorvo presentation
483	QORVO_00072556	QORVO_00072574	Qorvo presentation about data protection
484	QORVO_00072575	QORVO_00072591	Qorvo presentation about data protection
485	QORVO_00072592	QORVO_00072608	Qorvo presentation about data protection
486			Intentionally Omitted
487	QORVO_00073097	QORVO_00073102	Qorvo presentation about risk
488	QORVO_00074715	QORVO_00074733	Qorvo email exchange information sharing
489	QORVO_00074734	QORVO_00074750	Qorvo email exchange information sharing

DX	Beginning Bates	End Bates	Description
490	QORVO_00076715	QORVO_00076716	Internal Qorvo emails about Akoustis competition
491	QORVO_00076741	QORVO_00076742	Internal Qorvo emails about Akoustis competition
492	QORVO_00076749	QORVO_00076762	Internal Qorvo emails about filter strategy
493	QORVO_00077445	QORVO_00077447	Internal Qorvo emails about filter strategy
494	QORVO_00077687	QORVO_00077688	Internal Qorvo emails about Akoustis market share
495	QORVO_00077698	QORVO_00077701	Internal Qorvo emails about Akoustis
496	QORVO_00077711	QORVO_00077712	Internal Qorvo emails about Akoustis
497	QORVO_00077732	QORVO_00077766	Internal Qorvo emails about filters
498	QORVO_00077857	QORVO_00077862	Internal Qorvo emails about filters
499	QORVO_00077898	QORVO_00077902	Internal Qorvo emails about Akoustis vs Qorvo product development
500	QORVO_00077903	QORVO_00077907	Internal Qorvo emails about Akoustis vs Qorvo product development
501	QORVO_00077908	QORVO_00077912	Internal Qorvo emails about Akoustis vs Qorvo product development
502			Intentionally Omitted
503	QORVO_00079225	QORVO_00079242	Akoustis single crystal presentation
504			Intentionally Omitted
505	QORVO_00082434	QORVO_00082439	Qorvo emails about competition with Akoustis
506			Intentionally Omitted
507			Intentionally Omitted
508			Intentionally Omitted
509	QORVO_00101522	QORVO_00101533	Thomson Reuters Streetevents Edited Transcript of JP Morgan's 47 th Annual TMC Conference covering Qorvo dated 5/14/2019
510	QORVO_00102420	QORVO_00102487	Qorvo presentation on earnings (2020)
511	QORVO_00105527	QORVO_00105531	Internal Qorvo emails discussing losses to Akoustis
512	QORVO_00105928	QORVO_00105997	Qorvo presentation on earnings (2020)
513	QORVO_00106012	QORVO_00106079	Qorvo presentation on earnings (2020)

DX	Beginning Bates	End Bates	Description
514	QORVO_00108781	QORVO_00108850	Qorvo presentation on earnings (2020)
515			Intentionally Omitted
516	QORVO_00121551	QORVO_00121570	Slides from Qorvo presentation
517	QORVO_00121646	QORVO_00121648	Qorvo emails discussing comparison with Akoustis
518	QORVO_00121805	QORVO_00121819	Qorvo emails discussing band split filters
519			Intentionally Omitted
520			Intentionally Omitted
521	QORVO_00122488	QORVO_00122505	Qorvo emails about Netgear deal
522	QORVO_00122506	QORVO_00122507	Internal Qorvo emails about its BAW products
523			Intentionally Omitted
524	QORVO_00623975	QORVO_00624116	Qorvo presentation on quality
525	QORVO_00840830	QORVO_00840844	Qorvo's code of business conduct and ethics
526	QORVO_00842412	QORVO_00842415	Qorvo performance review of David Breton (2014-2015)
527	QORVO_00842697	QORVO_00842710	Qorvo performance review of Shawn Gibb (2019-2020)
528			Intentionally Omitted
529	QORVO_00627960	QORVO_00627960	Qorvo email about Akoustis
530	QORVO_00627966	QORVO_00627967	Qorvo email about Akoustis
531	QORVO_00627971	QORVO_00627971	Qorvo email about Akoustis
532	QORVO_00627973	QORVO_00627973	Qorvo email about Akoustis
533	QORVO_00636070	QORVO_00636076	Internal Qorvo email about chipset partners
534	QORVO_00637587	QORVO_00637687	Lengthy internal Qorvo email exchange about its product samples
535	QORVO_00637696	QORVO_00637819	Lengthy internal Qorvo email exchange about its product samples

DX	Beginning Bates	End Bates	Description
536	QORVO_00125500	QORVO_00125642	Qorvo presentation
537	QORVO_00126815	QORVO_00126816	Internal Qorvo emails about targeting Akoustis
538	QORVO_00127422	QORVO_00127425	Internal Qorvo emails about Akoustis filter
539	QORVO_00128068	QORVO_00128068	Internal Qorvo emails about pricing relative to Akoustis filter
540	QORVO_00128072	QORVO_00128072	Internal Qorvo emails about Akoustis procedures
541	QORVO_00133690	QORVO_00133708	Qorvo emails with Netgear about pricing
542			Intentionally Omitted
543	QORVO_00134266	QORVO_00134269	Internal Qorvo emails comparing Qorvo's products with Akoustis BAW filters
544	QORVO_00134307	QORVO_00134313	Internal Qorvo emails comparing Qorvo's products with Akoustis BAW filters
545	QORVO_00134461	QORVO_00134462	Internal Qorvo emails comparing Qorvo's products with Akoustis BAW filters
546			Intentionally Omitted
547	QORVO_00844195	QORVO_00844197	Internal Qorvo emails about discussions with Akoustis' R. Houlden
548	QORVO_00844198	QORVO_00844199	Emails between J. Shealy and Qorvo
549	QORVO_00844301	QORVO_00844304	White paper by Akoustis
550	QORVO_00844612	QORVO_00844619	Qorvo presentation titled "Qorvo's Response to Akoustis" (2016)
551	QORVO_00844658	QORVO_00844659	Internal Qorvo email about Akoustis paper (2016)
552	QORVO_00844710	QORVO_00844712	Internal Qorvo emails about publishing a paper to rebut Akoustis' claims
553	QORVO_00844713	QORVO_00844715	Qorvo whitepaper directed at Akoustis
554	QORVO_00844716	QORVO_00844717	Slides from Qorvo presentation
555			Intentionally Omitted
556	QORVO_00844826	QORVO_00844826	Internal Qorvo email about hiring Akoustis engineer

DX	Beginning Bates	End Bates	Description
557			Intentionally Omitted
558	AKTS_00075252	AKTS_00075253	Email from Rohan Houlden to Dave Aichele dated 12/2/2016 re: follow up from Aigner
559	AKTS_00091136	AKTS_00091272	U.S. Securities and Exchange Commission Form 10-K Annual Report Pursuant to Section 13 or 15(d) of the Securities Exchange Act of 1934 for Akoustis Technologies, Inc. for year ended June 30, 2019
560	AKTS_00116709	AKTS_00116711	Akoustis' proposal to Elta Systems Ltd. for development of BAW Discrete Filters dated 11/14/2016
561	AKTS_00194188	AKTS_00194191	Single-to-balanced Filters for Mobile Phone using Coupled Resonator BAW Technology article in 2004 IEEE Ultrasonics Symposium
562	AKTS_00280693	AKTS_00280712	Akoustis AIN transfer process with sacrificial layer presentation
563	AKTS_00396679	AKTS_00396689	Akoustis Wafer Exclusion Zone and Die Inspection Specification
564	AKTS_00612349	AKTS_00612349	Akoustis: Pattern trim overview presentation
565	AKTS_00641897	AKTS_00641897	Spreadsheet showing process flow for Akoustis ASML002
566	AKTS_00718958	AKTS_00719127	Deep Insight into Scandium Doped Aluminum Nitride FBAR_BAW Filter from Akoustis WiFi6 Connectivity Report dated 7/26/2021
567	AKTS_00779099	AKTS_00779148	MIL-STD_883H, Method 2010.12, Internal Visual (Monolithic) dated 2/26/2010
568	AKTS_00781763	AKTS_00781772	Joint JEDEC/ECIA/IPC Standard: Notification Standard for Product Discontinuance dated November 2014
569	AKTS_01199117	AKTS_01199123	STC Center of Excellence for Commercialization of MEMS & Smart System Technologies' presentation of "Calient Basebonding Module Inspection Document" dated 12/10/2013

DX	Beginning Bates	End Bates	Description
570	AKTS_01199177	AKTS_01199178	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Training Checklist Form for Theo Paran, Sr. Photo Engineer
571	AKTS_01199245	AKTS_01199258	Joint JEDEC/ECIA/IPC Standard: Customer Notification Standard for Product/Process Changes by Electronic Product Suppliers dated July 2014
572	AKTS_01199342	AKTS_01199343	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Phase Gate Review Record Form (blank)
573	AKTS_01199383	AKTS_01199384	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Design and Development Plan Form
574	AKTS_01199385	AKTS_01199387	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Procedure - Control of Design Changes dated 8/13/2012
575	AKTS_01199398	AKTS_01199401	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Process Deviation Waiver dated 9/17/2013
576	AKTS_01199402	AKTS_01199403	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering, Design Change Request Form
577	AKTS_01199405	AKTS_01199405	Design and development spreadsheet template
578	AKTS_01199422	AKTS_01199438	Akoustis Quality System Manual dated 2/17/2020
579	AKTS_01199439	AKTS_01199443	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Procedure - Design and Development dated 8/17/2012
580	AKTS_01199445	AKTS_01199445	Akoustis: Procedure - Design, Development and Maintenance dated 10/4/2021

DX	Beginning Bates	End Bates	Description
581	AKTS_01199622	AKTS_01199643	Akoustis: Policy - Quality Manual ISO 9001:2015 dated 9/21/2022
582	AKTS_01199668	AKTS_01199670	STC Smart System Technology & Commercialization Center, College of Nanoscale Science & Engineering: Operating Procedure - Changes to an Active Electronic Traveler dated 11/29/2011
583	AKTS_01199679	AKTS_01199680	Akoustis Branding Diagram for Document # AKF-1262-250 dated 11/17/2017
584	AKTS_01199872	AKTS_01199883	Akoustis Product Packaging specifications presentation dated Sept 2016
585	AKTS_01199981	AKTS_01200004	IPC/JEDEC J-STD-609B; Joint Industry Standard: Marking, Standards, and Labels of Leaded and Lead-Free Terminal Finished Materials Used in Electronic Assembly dated April 2016
586	AKTS_01200021	AKTS_01200026	Akoustis Die Assembly & Packaging presentation dated 12/4/2015
587	AKTS_01200027	AKTS_01200029	ASE (Chung-Li) Inc. Internal Marking Spec for Akoustis' AFK-1252-250 dated 6/12/2018
588	AKTS_01200093	AKTS_01200104	Kyocera: SMD Packages Division Presentation dated 6/30/2015
589	AKTS_01201440	AKTS_01201444	Akoustis: AKTS RF MTTF Method presentation (undated)
590	AKTS_01201862	AKTS_01201862	Akoustis: Activation Energy Update presentation dated 6/13/2018
591	AKTS_01201974	AKTS_01201980	Akoustis Corporate Presentation April 2017
592	AKTS_01204582	AKTS_01204582	Thamhain, H.J. (2000). Accelerating product developments via phase-gate processes. Paper presented at Project Management Institute Annual Seminars & Symposium, Houston, TX. Newtown Square, PA: Project Management Institute.
593	AKTS_01204583	AKTS_01204590	H. J. Thamhain's paper regarding "Accelerating product developments via phase-gate processes"

DX	Beginning Bates	End Bates	Description
594	AKTS_01204591	AKTS_01204596	BAW Resonator Design Considerations - An Overview (white paper) authored by Genrot G. Fattinger downloaded on 11/1/2023
595	AKTS_01204609	AKTS_01204591	Appropriate Methods to Suppress Spurious FBAR Modes in Volume Production dated 2006 downloaded on 11/1/2023
596	AKTS_01204613	AKTS_01204620	A Method for Accurate Modeling of BAW Filters at High Power Levels dated December 2016
597	AKTS_01204638	AKTS_01204751	JEDEC Publication: Failure Mechanisms and Models for Semiconductor Devices dated September 2016
598	AKTS_01204952	AKTS_01204969	JEDEC Publication: Guidelines for Visual Inspection and Control of Flip Chip Type Components (FCxGA) dated January 2013
599	AKTS_01204970	AKTS_01205031	JEDEC Publication: Guide to Standards and Publications Relating to Quality and Reliability of Electronic Hardware dated October 2013
600	AKTS_01205032	AKTS_01205113	JEDEC Standard: Inspection Criteria for Microelectronic Packages and Covers dated May 2017
601	AKTS_01205195	AKTS_01205214	JEDEC Standard: Semiconductor Wafer and Die Backside External Visual Inspection dated November 2021
602	AKTS_01205247	AKTS_01205264	JEDEC Standard: Temperature Cycling dated April 2023
603	AKTS_01205265	AKTS_01205284	JEDEC Publication: Characterization and Monitoring of Thermal Stress Test Oven Temperatures dated September 2019
604	AKTS_01205285	AKTS_01205324	JEDEC Standard: Standard Method for Measuring and Using the Temperature Coefficient of Resistance to Determine the Temperature of a Metallization Line dated September 2018
605	AKTS_01205325	AKTS_01205361	JEDEC Publication: Solid-State Reliability Assessment and Qualification Methodologies dated January 2019

DX	Beginning Bates	End Bates	Description
606	AKTS_01205362	AKTS_01205377	JEDEC Standard: External Visual dated April 2022
607	AKTS_01205378	AKTS_01205393	JEDEC Standard: Mark Legibility dated January 2020
608	AKTS_01205397	AKTS_01205410	JEDEC Standard: Temperature, Bias, and Operating Life dated November 2022
609	AKTS_01205411	AKTS_01206095	Department of Defense Test Method Standard Test Methods for Semiconductor Devices dated 11/20/2006
610	AKTS_01206096	AKTS_01206246	Performance Specification: Integrated Circuits (Microcircuits) Manufacturing, General Specification for dated December 1, 2002
611	AKTS_01206247	AKTS_01206328	Performance Specification: Hybrid Microcircuits, General Specification for dated January 6, 1999
612	AKTS_01206329	AKTS_01206352	JEDEC Standard: Reliability Qualification of Power Amplifier Modules dated March 2014
613	AKTS_01206353	AKTS_01206364	JEDEC Standard: Electrical Parameters Assessment dated October 2009
614	AKTS_01206365	AKTS_01206394	JEDEC Standard: Methods for Calculating Failure Rates in Units of Fits
615	AKTS_01206395	AKTS_01206699	Department of Defense: Test Method Standard: Electronic and Electrical Component Parts dated 6/28/2013
616	AKTS_01206700	AKTS_01206737	JEDEC Standard: Early Life Failure Rate Calculation Procedure for Semiconductor Components dated February 2007
617	AKTS_01206738	AKTS_01206767	JEDEC: Moisture/Reflow Sensitivity Classification for Non-hermetic Surface Mount Devices (SMDs) dated December 2022
618	AKTS_01206768	AKTS_01206780	EIA/JEDEC Standard: Standard for Failure Analysis Report Format dated December 1995
619	AKTS_01206781	AKTS_01206811	Joint Industry Standard: Handling, Packing, Shipping and Use of Moisture, Reflow, and Process Sensitive Devices dated April 2018

DX	Beginning Bates	End Bates	Description
620	AKTS_01206832	AKTS_01206851	JEDEC Standard: Method for Developing Acceleration Models for Electronic Device Failure Mechanisms dated 2022
621	AKTS_01206852	AKTS_01206861	Saw Duplexer Metallizations for High Power Durability dated 1998
622	AKTS_01206871	AKTS_01207532	Department of Defense: Test Method Standard Microcircuits dated 12/18/2000
623	AKTS_01210570	AKTS_01210657	Reverse Costing Analysis: Qorvo TQF6405 High Band FEM SMR-BAW Filter iPhone 6s Plus dated April 2016
624	AKTS_01210658	AKTS_01210658	Spreadsheet with a report/analysis of Apple iPhone 6S Plus Qorvo TQF-6405 FEM LTE High Band Reverse Costing dated March 2016 (NATIVE FILE)
625	AKTS_01210659	AKTS_01210961	System Plus Consulting: Reverse Costing - Structural, Process & Cost Report for Qorvo QM7810X 5G Sub-6 L-PAMiD w/Rx Div. dated October 2021
626	AKTS_01210962	AKTS_01210989	Tech Insights: Qorvo QPQ5500 Bulk Acoustic Wave (BAW) Band-Pass Filter dated 8/31/2023
627	AKTS_01210990	AKTS_01211002	Tech Insights: Circuit Vision Analysis on the Qorvo QPQ5500 BAW Filter dated 8/28/2023
628	AKTS_01211003	AKTS_01211039	Tech Insights: Qorvo QPQ5500 Bulk Acoustic Wave (BAW) Band-Pass Filter dated 8/29/2023
629	AKTS_01211040	AKTS_01211054	Tech Insights: Circuit Vision Analysis on the Qorvo QPQ5500 BAW Filter dated 8/29/2023
630	AKTS_01211055	AKTS_01211061	Tech Insights: Business Proposal: Revealing the innovations others cannot: Inside Advanced Technology Products dated 3/10/23
631	AKTS_01211062	AKTS_01211074	Tech Insights: Business Proposal: TechInsights is the most trusted source of actionable, in-depth intelligence to the semiconductor industry. Dated 7/27/2023
632	QORVO_00061075	QORVO_00061078	"BAW Filters for 5G Bands" article

Dx	Beginning Bates	End Bates	Description
633			Intentionally Omitted
634	QORVO_00844621	QORVO_00844621	White paper titled: High Power, Wideband Single Crystal XBAW Technology for sub-6 GHz Micro RF Filter Applications
635	N/A	N/A	Triquint Semiconductor, Inc. 885128; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 1/20/2016
636	N/A	N/A	Triquint Semiconductor, Inc. 885071-A; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 10/26/2015
637	N/A	N/A	Triquint Semiconductor, Inc. 885071; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 10/16/15
638	N/A	N/A	Triquint Semiconductor, Inc. 885070; 2437 MHz BAW filter dated 2/16/2015
639	N/A	N/A	Triquint Semiconductor, Inc. 885069; 2350 MHz BAW filter dated 3/3/2015
640	N/A	N/A	Triquint Semiconductor, Inc. 885067; 2.4GHz WLAN/BT LTE Co-Existence Filter dated 11/24/2014
641	N/A	N/A	Triquint 885062; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 2/10/2015
642	N/A	N/A	Triquint 885053; 1880 MHz BAW Filter dated 9/17/2014
643	N/A	N/A	Triquint Semiconductor, Inc. 885049; 2300-2400 MHz Tx/Rx Filter dated 10/20/2014
644	N/A	N/A	Triquint 885048; 1882.5 MHz BAW Filter dated 2/18/2015
645	N/A	N/A	Triquint 885033; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 12/22/14
646	N/A	N/A	Triquint Semiconductor, Inc. 885032; 2.4 GHz WLAN/BT LTE Co-Existence Filter dated 10/20/2014
647	N/A	N/A	Triquint 885026; 2595 MHz BAW Filter dated 4/9/2015
648	N/A	N/A	Triquint 885025; 1880 MHz BAW Filter dated 3/6/2014

Dx	Beginning Bates	End Bates	Description
649	N/A	N/A	Triquint 885024; 1960 MHz BAW Filter dated 11/21/2014
650	N/A	N/A	Triquint 885014; 2332.5 MHz BAW Filter dated 6/2/2014
651	N/A	N/A	Triquint 885010; 2442 GHz BAW Filter dated 2/18/2015
652	N/A	N/A	Triquint Semiconductor, Inc. 885009; 2535 MHz BAW Filter dated 9/20/2012
653	N/A	N/A	Triquint 885008; 2440 GHz BAW Filter dated 2/18/2015
654	N/A	N/A	Triquint Semiconductor, Inc. 885007; 2436 MHz BAW Filter dated 3/14/2011
655	N/A	N/A	Triquint TQQ7307: 2535 MHz LTE Band 7 Uplink BAW Filter dated 7/16/2014
656	N/A	N/A	Triquint TQQ7303: 1747.5 MHz LTE Band 3 Uplink BAW Filter Datasheet dated 7/16/2014
657	N/A	N/A	Qorvo TQQ7301: Band 1 Uplink BAW Filter Datasheet dated 2/15/16
658	N/A	N/A	Triquint TQQ0303: LTE Band 3 Downlink Module Datasheet dated 12/11/2014
659			Intentionally Omitted
660	N/A	N/A	Triquint TQQ0041: Band 41 Receive BAW Filter Datasheet Rev J dated 4/2/2014
661	N/A	N/A	Triquint Semiconductor, Inc. QPQ1291: Band 41 Tx/Rx Filter Preliminary Datasheet: Rev A dated 12/03/2015
662	N/A	N/A	Qorvo QPQ1290: Band 41 Tx/Rx Filter Datasheet dated 11/5/15
663	TRIQ2585240	TRIQ2585337	Avago Exhibit 234 to D.I. 389-10 dated 10/28/2011: Lunch & Learn BAW Technology at TriQuint dated 12/5/2008 (TRIQ2585240)
664	N/A	N/A	Electrical Characterization of Wafer Level Fan-Out (WLFO) Using Film Substrate for Low Cost Millimeter Wave Application dated 2010

DX	Beginning Bates	End Bates	Description
665	N/A	N/A	Ken-ya Hashimoto's "Chapter 8: BAW Piezoelectric Resonators" dated 2017
666	N/A	N/A	Akiyama's article "Enhancement of Piezoelectric Response in Scandium Aluminum Nitride Alloy Thin Films Prepared by Dual Reactive Cosputtering" dated 2009
667	N/A	N/A	Ramin Matloub's Research Article: "Electromechanical properties of $Al_{0.9}Sc_{0.1}N$ thin films evaluated at 2.5 GHz film bulk acoustic resonators" dated 8/30/2011
668	N/A	N/A	Milena Moreira's article "Aluminum scandium nitride thin-film bulk acoustic resonators for wide band applications" dated 3/29/2011
669	N/A	N/A	Robert Darveaux's article "Thermal and Power Cycling Limits of Plastic Ball Grid Array Assemblies" dated 1995
670	N/A	N/A	Robert Darveaux's article "Electromigration Performance of WLCSP Solder Joints" dated 2015
671	N/A	N/A	Robert Darveaux's PhD dissertation titled "Mechanical evaluation of indium for die attachment in a multichip package" dated 1993
672			Intentionally Omitted
673	AKTS_00073130	AKTS_00073133	Akoustis' Qorvo Visit Slides dated June 2016
674			Intentionally Omitted
675	AKTS_00201045	AKTS_00201054	Email from Hannah Stalvey to Dave Hodge dated 12/14/2015 re: Emailing: wallprint1, wallprint2, wallprint3, wallprint4, SCAR_Mask_5
676	AKTS_00278602	AKTS_00278610	Akoustis' Test Plan and Code Review Checklists Rev. A (undated)
677	AKTS_00593387	AKTS_00593538	Akoustis XBAW Technology for Connectivity dated July 2021 - version 2

DX	Beginning Bates	End Bates	Description
678	AKTS_00769521	AKTS_00769538	Akoustis' AIScN/GaN transfer structure and air bridge resonator process plus RF HEMT presentation (undated)
679	N/A	N/A	The Sedona Conference Commentary on Monetary Remedies in Trade Secret Litigation, Volume 24, dated 2023 (Ex. C to Bennis' Daubert Motion)
680			Intentionally Omitted
681	QORVO_00040320	QORVO_00040323	Mutual Non-Disclosure Agreement between Qorvo, Inc. and Coventor, Inc. dated 4/3/2015
682	QORVO_00634857	QORVO_00634864	Acoustic Filter Overview dated 2015
683	QORVO_00843706	QORVO_00843706	Mutual Non-Disclosure Agreement between TriQuint Semiconductor, Inc. and Symantec Corporation dated 8/12/2013
684			Intentionally Omitted
685			Intentionally Omitted
686	QORVO_00061700	QORVO_00061701	Email from Shawn Gibb to Gangadhar Burra dated 2/28/2019 re: RE: Welcome Shawn Gibb to Qorvo, HPS BU as Product Line Manager
687	QORVO_00066572	QORVO_00066574	Email from Cees Links to Leah Giovan dated 11/5/2018 re: RE: Akoustis
688	QORVO_00066585	QORVO_00066586	Email from Leah Giovan to Cees Links dated 11/5.2018 re: RE: Akoustis
689	QORVO_00076871	QORVO_00076871	Email from Shawn Gibb to Tuan Nguyen dated 11/8/2019 re: RE: You see the filters they claim being used?
690	QORVO_00105416	QORVO_00105417	Email from Brock West to Brock West dated 1/18/2021 re: RE: FY22 Wi-Fi Revenue review
691	QORVO_00109228	QORVO_00109229	Email from Brock West to Doug Delieto dated 9/4/2019 re: RE: AKTS manufacturing capabilities in nyc

DX	Beginning Bates	End Bates	Description
692	QORVO_00122459	QORVO_00122464	Email from John Anderton to Cees Links and Alex Zajac dated 9/16/2020 re: RE: IDP Eng Weekly Report - Filters - 2020 WW37
693			Intentionally Omitted
694	QORVO_00135187	QORVO_00135187	Email from Cees Links to Christian Lepiane and Roger Hall dated 2/7/2019 re: RE: Skyworks Solutions, Inc. (SWKS) CEO Liam Griffin o Q1 2019 Results - Earnings Call Transcript Seeking Alpha
695	QORVO_00172624	QORVO_00172625	Strategic Account - SWOT
696	QORVO_00073393	QORVO_00073399	Qorvo Policy Statement for handling certain data and selected security controls last dated 9/27/2020 (Robinson Depo. Ex. 9)
697	N/A	N/A	The Sedona Conference Commentary on Monetary Remedies in Trade Secret Litigation, a Project of The Sedona Conference Working Group (WG12) on Trade Secrets; May 2022 Public Comment Version (Bennis Depo. Ex. 4)
698	N/A	N/A	The Sedona Conference Journal article "Disputed Issues in Awarding Unjust Enrichment Damages in Trade Secret Cases"; Volume 19, Number 2 dated 2018 (Bennis Depo. Ex. 5)
699			Intentionally Omitted
700	AKTS_00440739	AKTS_00440753	Qorvo TGA2594-HM: 27 - 31 GHz GaN Power Amplifier Data sheet Rev. E, March 2021
701	AKTS_00440754	AKTS_00440757	Qorvo QM77038 Programming Update - Recommended Programming Sequence dated 1/20/2020

DX	Beginning Bates	End Bates	Description
702	AKTS_00440857	AKTS_00440864	Qorvo 885128: 2.4 GHz WLAN/BT LTE Co-Existence BAW Filter (Application Note) High-performance, high power Bulk Acoustic Wav (BAW) band-pass filter, high rejection in the band-edge and adjacent LTE/TD-LTE bands. Dated October 2020 Rev A.
703	AKTS_00440883	AKTS_00441221	Qorvo PAC22140 Battery Management System Product User Guide dated 2022
704	AKTS_00441222	AKTS_00441289	Qorvo ACT88760 Advanced PMIC with 7 Bucks, 6 LDOs Data Sheet Rev. B, 6/14/2022
705	AKTS_00441290	AKTS_00441363	Qorvo ACT88430 Advanced PMIC for Microcontrollers and Solid State Drive Applications Data Sheet Rev, G, August 2021
706	AKTS_00441406	AKTS_00441416	Qorvo QPL1811 75Ω CATV Amplifier 50 - 1800MHz Data Sheet - Rev. E, dated 8/25/2022
707	AKTS_00441417	AKTS_00441427	Qorvo QPL1811 75Ω CATV Amplifier 50 - 1800MHz Data Sheet - Rev. E, dated 8/25/2022
708	AKTS_00441428	AKTS_00441449	Qorvo QPL1163 75Ω 19dB CATV Amplifier (5-1218MHXz) Data Sheet - Rev B, dated 1/3/2023
709	AKTS_00441706	AKTS_00441721	Qorvo ACT4529 - 40 V/3.0 A CV/CC Buck Converter Featuring QC2.0, USB Auto-Detect and USB-PD Data Sheet Rev. F, dated February 2022
710	AKTS_00441748	AKTS_00441795	Qorvo Banned and Restricted Substances Specification SPE-001275, Rev F dated 9/30/2022
711	AKTS_00441796	AKTS_00441843	Qorvo Banned and Restricted Substances Specification SPE-001275, Rev F dated 9/30/2022
712	AKTS_00441858	AKTS_00441858	Qorvo's System for Award Management's Government Contract Information dated 2022
713	AKTS_00441859	AKTS_00441859	Qorvo's System for Award Management's Government Contract Information dated 2022

DX	Beginning Bates	End Bates	Description
714	AKTS_00441860	AKTS_00441866	Qorvo Onsite Service Provider Expectations for CSR Audits SPE-001698 Rev C dated 1/29/2021
715	AKTS_00441867	AKTS_00441868	Qorvo's Commitment to Ethical Labor Practices dated 7/7/2022
716	AKTS_00441869	AKTS_00441870	Qorvo Statement on CA Proposition 65 SPE-002105 Rev A dated 2/19/2020
717	AKTS_00441871	AKTS_00441872	Qorvo Statement on EU RoHS Directive SPE-002106 Rev A dated 2/19/2020
718	AKTS_00441875	AKTS_00441918	Qorvo Application Note for APS312 Production Tests for DW3000-Based Products dated June 2021 Rev A
719	AKTS_00458229	AKTS_00458240	Qorvo QPC7522 CATV SPDT Reflective Switch (5 - 3300MHz) Data Sheet Rev. E, dated 12/7/2020
720	AKTS_00132159	AKTS_00132160	Email from Jeffrey Shealy to Pauline Petrillo dated 11/13/2018 re: Re: From Leah
721	AKTS_00142738	AKTS_00142741	Process and Design Challenge for SMR-type Bulk Acoustic Wave (BAW) Filters at Frequencies Above 5 GHz article dated October 2019
722	AKTS_00212371	AKTS_00212392	IPC/JEDEC Joint Industry Standard: Moisture/Reflow Sensitivity Classification for Nonhermetic Solid State Surface Mount Devices dated March 2008
723	AKTS_00224690	AKTS_00224716	Qorvo Special Edition book titled "RF Filter Applications for dummies" authored by Larry Miller dated 2015
724	AKTS_00384715	AKTS_00384745	IPC/JEDEC Joint Industry Standard: Handling, Packing, Shipping and Use of Moisture, Reflow, and Process Sensitive Devices dated April 2018
725	AKTS_00434267	AKTS_00434319	2nd Qorvo Special Edition book titled "5G RF for dummies" dated 2020
726	AKTS_00440763	AKTS_00440764	Qorvo QM75041: 5G Power Amplifier Duplexer Module (PAMiD) Product Data Sheet Rev F dated 2023

Dx	Beginning Bates	End Bates	Description
727	AKTS_00440765	AKTS_00440766	Qorvo QM75041: 5G Power Amplifier Duplexer Module (PAMiD) Product Data Sheet Rev F dated 2023
728	AKTS_00440865	AKTS_00440869	Qorvo QPQ1904: Wi-Fi bandBoost BAW Filter Data Sheet dated 5/12/2020
729	AKTS_00440877	AKTS_00440878	Qorvo QPQ1904: 5 GHz UNII2c-3 bandBoost TM Filter Product Data Sheet dated 5/12/2020
730	AKTS_00440879	AKTS_00440880	Qorvo QPQ1903 Wi-Fi UNII1-2a bandBoost Filter Data Sheet Brief Rev A, dated November 2020
731	AKTS_00441367	AKTS_00441368	Qorvo Preliminary QPF4656 Wi-Fi 6E Front End Module Data Sheet Brief Rev A, dated February 2022
732	AKTS_00441369	AKTS_00441341	Qorvo QPF4656: 6 GHz Wi-Fi 6E High Power Front End Module Product Data Sheet Rev A - 02/2022
733	AKTS_00441372	AKTS_00441374	Qorvo QPF4632: 6 GHz Wi-Fi 6E Medium Power Front End Module Product Data Sheet Rev B dated 2/2022
734	AKTS_00441382	AKTS_00441384	Qorvo QPF4526: 5 GHz Wi-Fi Front End Module Product Data Sheet Rev B dated 11/2021
735	AKTS_00441385	AKTS_00441387	Qorvo QPF4526: 5 GHz Wi-Fi Front End Module Product Data Sheet Rev B dated 11/2021
736	AKTS_00441722	AKTS_00441723	Qorvo TQQ0041: 2593 MHz RF BAW Filter - Band 41 Rx Produce Data Sheet Rev K - dated 10/26/2017
737	AKTS_00441730	AKTS_00441731	Qorvo QM26003: Band 1/3/n41 Pentaplexer Product Data Sheet Rev C dated 2023
738	AKTS_00441732	AKTS_00441733	Qorvo QM26003: Band 1/3/n41 Pentaplexer Product Data Sheet Rev C dated 2023
739	AKTS_00441994	AKTS_00441994	Qorvo ESD Protection Circuits Product Data Sheet dated 2023
740	AKTS_00442533	AKTS_00442538	Article "Step Stress Accelerated Life Testing methods of lifetime predictions of SAW and BAW filters" dated October 2019

DX	Beginning Bates	End Bates	Description
741	AKTS_00452257	AKTS_00452350	U.S. Patent Shealy et al. Patent No.: US 10,985,732 B2 dated 4/20/2021 titled "5.6 GHZ Wi-Fi Acoustic Wave Resonator RF Filter Circuit"
742	AKTS_00452385	AKTS_00452479	U.S. Patent Shealy et al. Patent No.: US 10,979,023 B2 dated 4/13/2021 titled "5.9 GHZ C-V2X and DSRC Acoustic Wave Resonator RF Filter Circuit"
743	AKTS_00452480	AKTS_00452574	U.S. Patent Shealy et al. Patent No.: US 10,673,513 B2 dated 6/2/2020 titled "Front End Module for 5.2 GHZ WI-FI Acoustic Wave Resonator RF Filter Circuit"
744	AKTS_00452575	AKTS_00452671	U.S. Patent Shealy et al. Patent No.: US 11,063,576 B2 dated 7/13/2021 titled "Front End Module for 5.6 GHZ WI-FI Acoustic Wave Resonator RF Filter Circuit"
745	AKTS_00452672	AKTS_00452763	U.S. Patent Shealy et al. Patent No.: US 10,979,024 B2 dated 4/13/2021 titled "5.2 GHZ Wi-Fi Co-Existence Acoustic Wave Resonator RF Filter Circuit"
746	AKTS_00452764	AKTS_00452855	U.S. Patent Houlden et al. Patent No.: US 10,979,024 B2 dated 4/13/2021 titled "5G Band N79 Acoustic Wave Resonator RF Filter Circuit"
747	AKTS_00453143	AKTS_00453233	U.S. Patent Shealy Patent No.: US 11,316,496 B2 dated 4/26/2022 titled "Method and Structure for High Performance Resonance Circuit with Single Crystal Piezoelectric Capacitor Dielectric Material"
748	AKTS_00457650	AKTS_00457654	Qorvo QPQ1903 Wi-Fi bandBoost BAW Filter Data Sheet dated 5/15/2020
749	AKTS_00457712	AKTS_00457716	Mouser Electronics online order form for a Qorvo QPQ1904EVB01 dated 2023
750	AKTS_00457722	AKTS_00457722	Qorvo Process/Product Change Notification (PCN) dated 3/17/2017 changing "All commercial die sourced from Broomfield plant is obsolete due to plant closing."
751	AKTS_00457870	AKTS_00457888	Review of "Piezoelectric MEMS Resonators" book dated January 2017

DX	Beginning Bates	End Bates	Description
752	AKTS_00458138	AKTS_00458169	JEDEC Publication: Guideline for Switching Reliability Evaluation Procedures for Gallium Nitride Power Conversion Devices, Version 1.0, dated January 2021
753	AKTS_00458205	AKTS_00458206	Qorvo 880148: 2324 MHz RF BAW Filter Product Data Sheet dated 4/2007 (Wayback Machine webpage)
754	AKTS_00458213	AKTS_00458216	Qorvo 880157: 2560 MHz RF BAW Filter - S Band Product Data Sheet Rev B dated 12/2011 (Wayback Machine webpage)
755	AKTS_00458217	AKTS_00458219	Qorvo QPQ1029: Band 1 / Band 3 Dual BAW Filter Product Data Sheet Rev E dated 1/10/2020 (Product is obsolete as of 1/9/2023 - Wayback Machine webpage)
756	AKTS_00458220	AKTS_00458222	Qorvo QPQ1060: L1 Low Loss GPS SAW Filter Product Data Sheet Rev G dated 8/12/2022 (Wayback Machine webpage)
757	AKTS_00458245	AKTS_00458246	Qorvo online job opening for an Engineering Technician
758	AKTS_00458247	AKTS_00458252	High-temperature operating life Wikipedia page dated 7/25/2023
759	AKTS_00458308	AKTS_00458311	SEC Form 8-K for Qorvo, Inc. dated 5/15/2015
760	AKTS_00458312	AKTS_00458312	Qorvo online order form for QM45392: Front-End Module dated 2023
761	AKTS_00458313	AKTS_00458313	Qorvo online order form for QPQ1903EVB: RF Evaluation Board dated 2023
762	AKTS_00458393	AKTS_00458395	Qorvo SKY85746-11: 5 GHz 802.11ax Ultra-Linear WLAN Front-End Module with Integrated Log Detector (undated) (Skyworks webpage)
763	AKTS_00458396	AKTS_00458399	Qorvo SKY85780-11: WiFi 6E high-Power WLAN Front-End Module (undated) (Skyworks webpage)
764	AKTS_00458400	AKTS_00458402	Qorvo SKY85784-11: WiFi 6E high-Power WLAN Front-End Module (undated) (Skyworks webpage)
765	AKTS_00458436	AKTS_00459076	Department of Defense: Test Method Standard Microcircuits dated 12/31/1996

DX	Beginning Bates	End Bates	Description
766	AKTS_00459160	AKTS_00459164	JEDEC Wikipedia webpage dated 12/20/2022
767	AKTS_00459185	AKTS_00459188	MIL-STD-883 Wikipedia webpage dated 7/26/2023
768	AKTS_00459189	AKTS_00459480	"RF Bulk Acoustic Wave Filters for Communications" book dated 2009
769	AKTS_00465305	AKTS_00465328	IPC/JEDEC Joint Industry Standard: Moisture/Reflow Sensitivity Classification for Nonhermetic Surface Mount Devices dated December 2014
770	AKTS_00483778	AKTS_00483793	JEDEC Standard: Preconditioning of Nonhermetic Surface Mount Devices Prior to Reliability Testing dated August 2003
771	AKTS_00545124	AKTS_00545151	JEDEC Publication: Long Term Storage for Electronic Solid-State Wafers, Dice, and Devices dated August 2022
772	AKTS_00606577	AKTS_00606578	Qorvo QPQ190-3: 5 GHz UNII1-2a bandBoost Filter Product Data Sheet dated 2020 (Wayback Machine webpage)
773	AKTS_00606581	AKTS_00606582	Qorvo QPQ1904: 5 GHz UNII2c-3 bandBoost TM Filter Product Data Sheet dated 5/12/2020 (Wayback Machine webpage)
774	AKTS_00682214	AKTS_00682216	Qorvo QPQ 1285: 2503 MHz RF BAW Filter - Band 41 Tx/Rx End of Life Announced 7/26/2022 from Qorvo 2023 webpage
775	AKTS_00693733	AKTS_006893736	"BAW Filters for 5G Bands" article authored by R. Aigner, et al. dated 2018
776	AKTS_00755473	AKTS_00755473	Mouser Electronics online order form for a Qorvo QPQ1904EVB01 dated 2023
777	AKTS_00755474	AKTS_00755475	Qorvo QPQ1904: 5 GHz UNII2c-3 bandBoost TM Filter (Qorvo webpage dated 8/14/2023)
778	AKTS_00757119	AKTS_00757146	Joint Industry Standard: Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices dated February 2012

DX	Beginning Bates	End Bates	Description
779	AKTS_00761215	AKTS_00761222	Qorvo 885136: 2437 MHz BAW Filter Data Sheet: Rev F dated May 2017
780	AKTS_00764391	AKTS_00764421	Joint Industry Standard: Handling, Packing, Shipping and Use of Moisture, Reflow, and Process Sensitive Devices dated April 2018
781	AKTS_00778932	AKTS_00778933	Qorvo QPQ1904: 5 GHz UNII2c-3 bandBoost™ Filter (Qorvo webpage dated 8/14/2023)
782			Intentionally Omitted
783			Intentionally Omitted
784	AKTS_00778950	AKTS_00778950	Spec. sheet for Qorvo's QPF4632 product from Qorvo's website.
785	AKTS_00779017	AKTS_00779019	Qorvo QPF4632: 6 GHz Wi-Fi 6E Medium Power Front End Module Product Data Sheet Rev B - dated 02/2022 (Qorvo webpage dated 8/14/2023)
786	AKTS_00779051	AKTS_00779053	TriQuint Semiconductor RFFM4252 Datasheet from ALLDATASHEET.COM website dated 8/14/2023
787	AKTS_00779093	AKTS_00779098	"Calculating Reliability using FIT & MTTF; Arrhenius HTOL Model" article by Paul Elleman dated 1/9/2012 copyright © Microsemi Corp.
788	AKTS_00779156	AKTS_00779156	Native file - ResearchGate webpage for a publication titled "Piezoelectric MEMS Resonator," dated January 2017.
789	AKTS_00779342	AKTS_00779530	scia systems "Software Manual scia Trim 200" dated December 2017
790	AKTS_00779536	AKTS_00779544	scia systems "Quick Guide: Backup and Restore Axis Controller" dated March 2019
791	AKTS_00779589	AKTS_00779683	scia systems "Maintenance Manual scia Trim 200" dated December 2017
792	AKTS_00779791	AKTS_00779911	scia systems "Instruction Handbook scia Trim 200" dated December 2017
793	AKTS_00779935	AKTS_00780001	scia systems "Instruction Handbook Ion Beam Source" RF37-e/RF37-I dated Mai 2017

Dx	Beginning Bates	End Bates	Description
794	AKTS_00780001	AKTS_00780095	scia systems "Quick Guide: Replace the Plasma Cup RF37-3/RF-I decentral " RF37e/RF37-I decentral dated May 2019
795	AKTS_00780210	AKTS_00780211	scia systems "Adjustment of Phi-Axis limit switch and 0° position" (undated)
796	AKTS_00780212	AKTS_00780214	scia systems "Cleaning of the Chuck O-Ring"
797	AKTS_00780248	AKTS_00780313	scia systems "Instruction Handbook scia Trim 200" dated March 2019
798	AKTS_00780336	AKTS_00780345	scia systems "Guidance: Ignition Generator: Exchange Circuit Board" dated December 2017
799	AKTS_01030644	AKTS_01030648	Qorvo QPQ1904: Wi-Fi bandBoost BAW Filter Data Sheet dated 5/12/2020
800	AKTS_01134048	AKTS_01134048	Qorvo online order form for QPQ1903EVB: RF Evaluation Board dated 10/19/2023
801	AKTS_01134504	AKTS_01134506	TriQuint Semiconductor RFFM4252 Datasheet from ALLDATASHEET.COM website (undated)
802	AKTS_01134508	AKTS_01134529	Qorvo Store: Qorvo 1903EVB, Evaluation Board from Qorvo (undated)
803	AKTS_01134531	AKTS_01134552	Qorvo Store: Qorvo QM45392: Front-End Module from Qorvo (undated)
804	AKTS_01134577	AKTS_01134592	Qorvo Store: Qorvo QM45392: Front-End Module from Qorvo (undated)
805	AKTS_01134609	AKTS_01134609	ALLDATASHEET.COM webpage of Electronics Components Datasheet Search of Qorvo RFFM4252 Wi-Fi Front End Module (undated)
806	AKTS_01134612	AKTS_01134612	Mouser Electronics webpage for Qorvo RFSW8009TR7 dated 2023
807	AKTS_01134701	AKTS_01134710	JEDEC Standard: High Temperature Storage Life dated December 2010
808			Intentionally Omitted
809	AKTS_01134711	AKTS_01134724	JEDEC Standard: Electrostatic Discharge (ESD) Sensitivity Testing, Machine Model (MM) dated November 2010

DX	Beginning Bates	End Bates	Description
810	AKTS_01134725	AKTS_01134742	JEDEC Standard: Temperature Cycling dated March 2009
811	AKTS_01134743	AKTS_01134754	JEDEC Standard: Power and Temperature Cycling dated January 2004, Reaffirmed 2011
812	AKTS_01134755	AKTS_01134768	JEDEC Standard: Temperature, Bias, and Operating Life dated November 2010
813	AKTS_01134769	AKTS_01134776	JEDEC Standard: Physical Dimensions dated June 2003 (Reaffirmed: June 2006)
814	AKTS_01134777	AKTS_01134802	JEDEC Standard: Stress-Test-Driven Qualification of Integrated Circuits dated March 2009
815	AKTS_01134803	AKTS_01134856	ANSI/ESDA/JEDEC JS-001-2012: For Electrostatic Discharge Sensitivity Testing: Human Body Model (HBM) - Component Level dated April 2012
816	AKTS_01134857	AKTS_01134874	JEDEC Standard: Wire Bond Shear Test Method dated August 2009
817	QORVO_00135196	QORVO_00135196	Email from Eid Alsabbagh to Tony Testa dated 12/23/2019 re: RE: 5.2GHz/5.6GHz filters (1903/1904)
818	QORVO_00166534	QORVO_00166537	Email from Scott Muir to John Anderton, Alex Zajac, and Wayne Polonio dated 1/20/2020 re: RE: QPQ1904 MRD and PRD
819	QORVO_00166568	QORVO_00166570	Email from Bettina Meles to Greg Caruyer dated 12/24/2019 re: RE: QPQ104 timeline
820	QORVO_00167310	QORVO_00167312	Email from John Anderton to Wireless.Connectivity.PLM dated 6/25/2019 re: FW: Ruckus Wireless Mtg Notes: June 12, 2019
821	QORVO_00172672	QORVO_00172673	Email from Wayne Polonio to Jeff Mink and Kevin Bogacki dated 4/27/2018 re: Re: 5GHz BAW
822	QORVO_00172803	QORVO_00172805	Email from Tony Testa to Tommy Tran and Jeff Lin dated 6/17/2020 re: RE: Qorvo/Netgear - RZX70 - 5G BAW Diving Catch opp - Akoustis Delayed 2.4Mpcs

DX	Beginning Bates	End Bates	Description
823	QORVO_00173567	QORVO_00173567	Email from Wayne Polonio to Kathy DaSilva, Bror Peterson, and Chris Levesque dated 7/1/2021 re: Charter Quadband & Broadcom
824	QORVO_00173590	QORVO_00173592	Email from Greg Caruyer to Cees Links, Tony Testa, Wayne Polonio, and Bettina Meles dated 1/17/2020 re: RE: QPQ1904 timeline
825	QORVO_00174244	QORVO_00174254	Email from Wayne Polonio to Nathan Chen, David Guo, Paul Butler, Jaidev Sharma, Jeff Lin, and Elba BAI dated 3/1/2021 re: Re: QPQ1907
826	QORVO_00174415	QORVO_00174415	Email from Jeremy Foland to Wayne Polonio, Igor Lalicevic, Jeff Lin, and Tony Testa dated 6/8/2015 (no subject)
827			Intentionally Omitted
828	QORVO_00174528	QORVO_00174537	Email from Kevin Bogacki to Pat Clancy dated 2/7/2020 re: RE: Qorvo/EERO/Eden - QPF4588A
829	QORVO_00305637	QORVO_00305651	Qorvo PDE Doc Center Training: Submitting Documents for IDP Day P rollout dated 7/14/2016
830	QORVO_00308585	QORVO_00308586	Email from Larry Blackledge to Hans Schwarz dated 11/9/2019 re: RE: OPCP AKTS (OUTPERFORM) - Multiple Products on the Horizon
831			Intentionally Omitted
832	QORVO_00309117	QORVO_00309133	Qorvo FY 20 Enterprise Risk Assessment Interviews dated 7/11/2019
833	QORVO_00311443	QORVO_00311443	Email from Bob Bruggeworth to Hans Schwarz dated 1/21/2019 re: FW: Akoustis
834	QORVO_00311755	QORVO_00311755	Email from Bob Bruggeworth to Mark Murphy dated 11/14/2018 re: Re: talked John Kurtzwell
835	QORVO_00512164	QORVO_00512215	Qorvo Competitive Analysis Update: 2018 QTEQ Ken Schroeter dated 2015

Dx	Beginning Bates	End Bates	Description
836	QORVO_00512639	QORVO_00512656	Email from Terry Glascock to Rufus_Chen@aseglobal.com dated 8/3/2018 re: RE: BAW with WLFO (Security B)
837	QORVO_00512653	QORVO_00512655	Qorvo BAW Terry Glascock dated 8/3/2018
838	QORVO_00512656	QORVO_00512656	ASE Spreadsheet of design process requirements
839	QORVO_00512673	QORVO_00512673	ASE Spreadsheet of design process requirements
840	QORVO_00512677	QORVO_00512692	Email from Terry Glascock to Rufus_Chen@aseglobal.com dated 8/3/2018 re: RE: BAW with WLFO (Security B)
841	QORVO_00512694	QORVO_00512711	Email Rufus_Chen@aseglobal.com to Terry Glascock dated 8/7/2018 re: RE: BAW with WLFO (Security B)
842	QORVO_00512713	QORVO_00512727	Email from Terry Glascock to Rufus_Chen@aseglobal.com and Steve Bosworth dated 8/10/2018 re: RE: FW: BAW with WLFO (Security B)
843	QORVO_00518367	QORVO_00518539	BAW Filter Comparison 2022 presentation
844			Intentionally Omitted
845	QORVO_00518573	QORVO_00518581	Agenda dated 2023
846	QORVO_00519605	QORVO_00519606	Email from Kevin Schoenrock to Wayne Polonio, Tony Testa, and Bob Baeten dated 2/26/2023 re: RE: AKTS filter performance feedback
847	QORVO_00520030	QORVO_00520031	Email to Doug Delieto (undated) re: Action items
848			Intentionally Omitted
849	QORVO_00531612	QORVO_00531620	Qorvo Agenda, not dated
850	QORVO_00538879	QORVO_00538879	Spreadsheet - Themes of Concerns/Staff call 2/16/18
851	QORVO_00539180	QORVO_00539180	Spreadsheet - Themes of Concerns/Staff call 2/16/18
852			Intentionally Omitted
853	QORVO_00557144	QORVO_00557145	Email from Kevin Schoenrock to Doug Delieto dated 3/3/23 re: Action items

DX	Beginning Bates	End Bates	Description
854	QORVO_00578993	QORVO_00579013	Edited 8/2/17 Transcript - Q1 2017 Qorvo Inc Earnings Call
855	QORVO_00596746	QORVO_00596812	Mobile Experts Semiconductors for Remote Radio Heads 2022, August 2022
856	QORVO_00602923	QORVO_00602924	Email from Jeff Jones to Kevin Schoenrock et al. dated 2/25/23 re: CPE Wi-Fi Filter Status based on field input, with attached Competitive Filter Performance Summary 9-14-22
857	QORVO_00647897	QORVO_00647897	Broadcom Power/EVM graph
858	QORVO_00647935	QORVO_00647941	Email from Vincent Lin to Wayne Polonio dated 3/31/15 re: The latest chipset partners update
859	QORVO_00651150	QORVO_00651151	Email from Eid Alsabbagh to Rohan Houlden et al dated 4/29/15 re: Skywork parts discussed by customers on the trip to Taiwan
860	QORVO_00651152	QORVO_00651168	Skyworks Preliminary Data Sheet SKY85806-Dual-Band 802.11 a/g/n/ac Wireless LAN Front-End
861	QORVO_00651169	QORVO_00651180	Skyworks Preliminary Data Sheet SKY85728-11: 5 GHz WLAN Front-End Module
862	QORVO_00651192	QORVO_00651263	Skyworks 802.11ac Product Roadmap April 2015
863	QORVO_00651272	QORVO_00651292	Microsemi RF Wireless Technology Solutions Feb 2014 Update
864	QORVO_00652196	QORVO_00652199	Email from Wayne Polonio to Tony Testa and John Anderton dated 2/17/20 re: EERO-5GHz BAW
865	QORVO_00652397	QORVO_00652401	Email from Tommy Tran to Tony Testa et al. dated 6/17/20 re: Plume 1903/1904-Augustus
866	QORVO_00652439	QORVO_00652439	Email from Cees Links from Tony Testa dated 7/21/20 re: EPS call
867	QORVO_00652968	QORVO_00652970	Products information of 5.49 GHz BPF
868	QORVO_00653675	QORVO_00653677	Email from Brock West to Tony Testa and Doug Delieto dated 9/4/19 re: FYI: Oppenheimer on AKTS

DX	Beginning Bates	End Bates	Description
869	QORVO_00664990	QORVO_00664990	Email from Ahmad Abdelmajid to Bob Baeten et al. dated 1/18/16 re: Emailing - SKY85410 v SKY85405 v RFPA5542 Comparison.pdf
870	QORVO_00664991	QORVO_00664998	Skyworks SKY85410-11, 5GHz 23dBm 802.11ac PA, Log Det
871	QORVO_00667049	QORVO_00667049	Email from Wilco Van Hoogstraeten to Alex Zajac et al dated 3/3/17 re: AKTS: Two Customers Signed and More in the Works
872	QORVO_00676705	QORVO_00676721	Qorvo Executive S&OP - Dec '20 Meeting Summary, 12/11/20
873	QORVO_00689001	QORVO_00689014	Email from Teresa Lai to Jop Voorhoeve and John Anderton dated 1/8/2020 re: RE: Gross Margin Improvements - pricing
874	QORVO_00694741	QORVO_00694741	Email from Tony Testa to Cheng-hui Lin et al. dated 7/14/16 re: SKY new 5G 3x5mm2 HP FEM-SKY85743-11
875	QORVO_00696867	QORVO_00696869	Email from Cheng-Hui Lin to Pat Clancy et al. dated 2/16/16 re: [Competitive Analysis] - SKY85728/31 5GHz 3X5mm FEM
876	QORVO_00706697	QORVO_00706698	Email from Corrine Chen to Tommy Tran re: Netgear R7000-SE5003 Price?
877	QORVO_00706699	QORVO_00706710	Email from Cheng-Hui Lin to Maurice Huang et al. dated 3/19/2015 re: RE: Netgear - Qorvo 5G PA - SW Assistance NightHawk R7000 Benchmark
878	QORVO_00707312	QORVO_00707326	Email from Tommy Tran to Wayne Polonio dated 2/11/20 re: EERO Eden samples for Feb-Confirm ES2-QPF4558A-Need Approval before Shipping
879	QORVO_00719053	QORVO_00719061	Qorvo PPT Template, Tips and Best Practices
880	QORVO_00745771	QORVO_00745794	Qorvo Competitive Strategy - How to Compete with AVGO, How Will We Win?, Design for Speed
881	QORVO_00749329	QORVO_00749330	Email from Kevin Schoenrock to Jeremy Foland dated 2/24/23 re: Plume: Discussion with Jeff Liew today

DX	Beginning Bates	End Bates	Description
882	QORVO_00749331	QORVO_00749332	Email from Kevin Schoenrock to Jeremy Foland dated 2/24/23 re: Plume: Discussion with Jeff Liew today
883	QORVO_00751914	QORVO_00751915	Email from Bob Baeten to Ahmad Abdelmajid dated 6/24/19 re: Ruckus Wireless Mtg Notes: June 12, 2019
884	QORVO_00752106	QORVO_00752106	Email from Ahmad Abdelmajid to Cheng-Hui Lin dated 1/11/19 re: Akoustis Small-Signal Data
885	QORVO_00752473	QORVO_00752476	Email from Anthony Testa Jr. to Vincent Vermeer et al. dated 2/7/19 re: Conversation with Anthony Testa Jr. Eid Alsabbagh, Igor Lalicevic, Jaidev Sharma, Jervais Seegars, Vincent Vermeer
886	QORVO_00753641	QORVO_00753652	Skyworks Preliminary Data Sheet, SKY85310-11: 2.4 GHz, 356 QAM WLAN/Bluetooth Front-End Module
887	QORVO_00773812	QORVO_00773830	Qorvo QPB1034 S-Band Switched Filter Bank Module MRD, 3/3/23
888	QORVO_00794273	QORVO_00794284	Qualcomm WCN3990 RD2 Chain 1 WLAN Front End Module Mini-Specification, 6/2/16
889	QORVO_00799560	QORVO_00799560	Email from Michael Morgensen to Jonathan Robbins dated 4/20/16 re: waveform sharing with Keysight
890	QORVO_00799561	QORVO_00799571	Qorvo WLAN measurement discussion with Keysight, 4/19/16
891	QORVO_00804547	QORVO_00804562	Email from Kent Leverich to Jie Hu dated 8/12/21 re: Leverich Weekly Update
892	QORVO_00808248	QORVO_00808250	Email from Kevin Bogacki to Bernard de Neeve dated 11/30/16 re: Akoustis new investor
893	QORVO_00811723	QORVO_00811733	Skyworks Preliminary Data Sheet, SKY85717-11: 5 GHz, 802.11ac Front-End Module
894	QORVO_00815352	QORVO_00815419	Qorvo IDP Earnings Prep Session (EPS), 7/20/20
895	QORVO_00820441	QORVO_00820636	Report - RF Front Ends for Mobile Devices 2019

DX	Beginning Bates	End Bates	Description
896	QORVO_00837177	QORVO_00837177	Email from Jean Briot to Berry Leonard dated 5/8/17 re: FW: Filter Attrition
897	QORVO_00845698	QORVO_00845699	Email from Robert Aigner to Otto Berger and Gernot Fattinger dated 3/7/16 re: IDP 5GHz
898	QORVO_00846971	QORVO_00846974	Email from Eid Alsabbagh to Robert Aigner dated 11/10/15 re: Reference Filtre a cavite sur U-Big
899	QORVO_00849665	QORVO_00849692	Email from Scott Muir to Kevin Schoenrock, Jack Underwood, Mudar Al-Joumayly, Jaidev Sharma, Nathan Chen, and Jeff Mink dated 8/22/23 re: RE: AKST Competitive S Params attaching Avago Preliminary DataSheet, ACPF-W065, 6GHz WiFi Bandpass Filter for Coexistence with 5GHz WiFi
900	AKTS_00037311	AKTS_00037323	Akoustis presentation titled "Trim System Evaluation Plan"
901	AKTS_00037339	AKTS_00037339	Trim Work Center Update (undated)
902	AKTS_00050552	AKTS_00050582	Akoustis' Resonator, Filter Design & Modeling dated 11/4/2019
903	AKTS_00076554	AKTS_00076556	Akoustis Proposed Trimming Options (undated)
904	AKTS_00194068	AKTS_00194068	MSET082-XB1, 3.0-3.5 GHz, DE: Saurabh, Layout: David, AKF10132-6 (undated)
905			Intentionally Omitted
906			Intentionally Omitted
907			Intentionally Omitted
908	AKTS_01039458	AKTS_01039627	Akoustis XBAW Technology for Connectivity dated July 2021 - version 2
909			Intentionally Omitted
910	QORVO_00653743	QORVO_00653744	Email from Wilco Van Hoogstraeten to Cees Links, Tony Testa, and Ahmad Abdelmajid dated 2/11/2019 re: RE: AKTS

DX	Beginning Bates	End Bates	Description
911			Intentionally Omitted
912			Intentionally Omitted
913			Intentionally Omitted
914	QORVO_00000008	QORVO_00000284	USPTO Patent Number: 7522018, Issue Dated: 4/21/2009; Title of Invention: Electro-Acoustic Resonator
915	NGUYEN_000009	NGUYEN_000024	An Integrated CMOS Micromechanical Resonator High-Q Oscillator article dated April 1999
916	NGUYEN_000025	NGUYEN_000035	Evaluation of Titanium as a Diffusion Barrier Between Aluminum and Silicon for 1.2 μm CMOS Integrated Circuits dated November 1987
917	NGUYEN_000036	NGUYEN_000039	Precision Frequency Control Volume 1 Acoustic Resonators and Filters dated 1985
918	NGUYEN_000040	NGUYEN_000052	Trapped-Energy Modes in Quartz Filter Crystals dated 9/14/1966
919	NGUYEN_000053	NGUYEN_000076	U.S. Patent No. 4,135,108 dated 1/16/1979 titled "Quartz Resonator with the Electrodes that do not Adhere to the Crystal, Author: Raymond J. Besson
920	NGUYEN_000077	NGUYEN_000085	Handbook of Filter Syntheses dated 1967
921	AKTS_00036779	AKTS_00036779	Email from Dave Aichele to Dave Hodge dated 9/24/2017 re: 5.2GHz S-Part
922			Intentionally Omitted
923	AKTS_00192533	AKTS_00192533	Native File: Akoustis Technologies, Inc.'s 5 Year Strategic Plan - Consolidated Income Statement dated 6/30/2022
924	AKTS_00192534	AKTS_00192554	Akoustis "Darpa: Competitive Landscape and Techno-Economic Analysis" presentation dated 6/30/2022

DX	Beginning Bates	End Bates	Description
925	AKTS_00432620	AKTS_00432635	Akoustis - Phase I - Interim Report: Summary of Phase I Activities dated July 2015
926	AKTS_00433848	AKTS_00433867	SBIR/STTR Report Cover Page (10-17-2012); NSF Award Number: 1556097; Project Title: SBIR Phase II - Single Crystal Group III-N Bulk Acoustic Resonators and Bulk Acoustic Wave Filter Components for Mobile Communications
927	AKTS_00433875	AKTS_00433890	Akoustis "Single Crystal Group III-Nitride Bulk Acoustic Resonators and BAW Filter Components for Mobile Communications" SBIR Phase 1 Award #1447854; Principle Investigator: Dr. Jeffrey Shealy; Update: 1/26/2015
928	AKTS_01200304	AKTS_01200304	Native file: Akoustis FY24 Forecast 20231101 BU
929	AKTS_01211075	AKTS_01211075	2023 W-2 and Earnings Summary for Antonio Espinoza
930	QORVO_00024338	QORVO_00024343	Akoustis Technologies Completes Strategic Acquisition of New York Wafer Manufacturing Facility and Operation dated 6/27/2017
931			Intentionally Omitted
932	QORVO_00610960	QORVO_00610966	Page Vault - Akoustis' Release: Akoustis® Technologies Ships First BAW Filter Prototypes Operating above 3.5GHz: Akoustis Technologies, Inc. (AKTS)
933	N/A	N/A	Fine Pitch Copper Pillar Flip Chip, Amkor (June 17, 2017) at The Wayback Machine
934	N/A	N/A	Latest OIDA roadmap on quantum photonics: (https://opg.optica.org/abstract.cfm?URI=OIDA-2020-3)
935	N/A	N/A	<u>LEDs Magazine webpage: Latest OIDA roadmap on quantum photonics:</u> (https://opg.optica.org/abstract.cfm?URI=OIDA-2020-3)

DX	Beginning Bates	End Bates	Description
936	N/A	N/A	<u>LIGHTWAVELOGIC webpage:</u> <u>https://www.lightwavelogic.com/wp-content/uploads/sites/319/2021/03/WP_IPS_R-Roadmap-Decoded.pdf</u>
937	N/A	N/A	“Closed-Form Expressions for the Design of Ladder-Typer FBAR Filters” article by Óscar Menéndez, <i>Member IEEE</i> , Pedro de Paco, Ramón Villarino, <i>Member IEEE</i> , and Josep Parró dated December 2000
938	N/A	N/A	"Wideband Bandpass Filters with SAW-Filter-Like Selectivity Using Chip SAW Resonators" article by Xiaoming Lu, <i>Student Member, IEEE</i> , Koen Mouthaan, <i>Member, IEEE</i> , and Yeo Tat Soon, <i>Fellow, IEEE</i> dated January 2014
939	N/A	N/A	“Analysis of Thermal Vias in High Density Interconnect Technology” article by S. Lee, T.F. Lemczyk, and M.M. Yovanovich dated February 1992
940	N/A	N/A	Email from Photonic Integrated Circuits conference to drmlabby@gmail.com (Michael Lebbby) dated 3/9/2016 re: Last chance to download the conference presentations (Exhibit 1)
941	N/A	N/A	Email from Michael Lebbby to Adam Richardson dated 2/7/2018 re: Re: PIC International: UPDATED Presentation Guidelines & Deadlines Reminders (Exhibit 2)
942	N/A	N/A	Qorvo presentation titled “The Quest for 5G CS International 2016” (Exhibit 3)
943	N/A	N/A	Organic versus Compound Semiconductor; Competitive or Complementary Materials for Particular Applications dated 2016 presentation (Exhibit 4)
944	N/A	N/A	Comparing GaAs and GaN technologies for RF dated July 2017 with presentation attached titled "Comparing GaAs and GaN technologies for RF" dated 7/24/2017

DX	Beginning Bates	End Bates	Description
945	N/A	N/A	Qorvo "Comparison of FOWLP vs QFN package from Thermal Aspect" presentation dated 3/17/2015
946	N/A	N/A	J. Kosinski, A. Ballato, S. Mallikarjun, "Quartz Crystal Resonator Mass Loading," Research and Development Technical Report, Electronics Technology and Devices Laboratory, U.S. Army Laboratory Command, Fort Monmouth, New Jersey, Aug. 1989
947			Intentionally Omitted
948	AKTS_00076584	AKTS_00076588	Akoustis 5GHz UNI Specs & Sample Timeline Aug 2017
949	AKTS_00086138	AKTS_00086203	Email from Rohan Houlden to Jeffrey Shealy dated 3/22/2019 re: FW: Executive Weekly Update: Product Development Status 3/22
950	AKTS_00102341	AKTS_00102341	01222021 Akoustis Products Design Spec List
951	AKTS_00102342	AKTS_00102349	Akoustis A10160 - 6 GHz WiFi 6E Coexistence BAW Filter dated 12282020
952	AKTS_00102350	AKTS_00102357	Akoustis A10158 - 5.8 GHz WiFi 6E Coexistence BAW Filter dated 12282020
953	AKTS_00102358	AKTS_00102365	Akoustis A10154 - 5.4 GHz WiFi 6E Coexistence BAW Filter dated 12282020
954	AKTS_00119934	AKTS_00119967	Akoustis Board of Directors Meeting November 2017 Presentation
955	AKTS_00612271	AKTS_00612277	Akoustis Board of Directors Meeting February 2018 Presentation
956	AKTS_00762969	AKTS_00762986	Akoustis Board of Directors Meeting February 2018 Presentation
957	N/A	N/A	Forensic & Valuation Services Practice Aid: Attaining Reasonable Certainty in Economic Damages Calculations," American Institute of Certified Public Accountants, 2020 ("AICPA Practice Aid (2020)")

DX	Beginning Bates	End Bates	Description
958	N/A	N/A	Robin Erickson, “Calculating the True Cost of Voluntary Turnover: The Surprising ROI of Retention,” Deloitte Research Bulletin, 2016 (“Erickson (2016)”)
959	N/A	N/A	Roman L. Weil, et al., <i>Litigation Services Handbook: The Role of the Financial Expert</i> , 6th ed. (New York: John Wiley & Sons, Inc., 2017) (“Litigation Services Handbook”) – Chapter 4, p. 3, and Chapter 4.6
960	N/A	N/A	National Research Council, Reference Manual on Scientific Evidence, 3rd ed. (Washington, DC: The National Academies Press, 2011) (“Reference Manual on Scientific Evidence”) – p. 433
961	N/A	N/A	Internet Archive authenticated copies of DTX0729, DTX0730, DTX0753-DTX0756, DTX0933
962	N/A	N/A	Internet Archive authenticated copies of DTX0635-DTX0657, DTX0660-DTX0662
963	AKTS_00155055	AKTS_00155066	Email from Dave Hodge to Frank Bi and Rohan Houlden dated 12/7/2020 re: RE: ASML007, A10252
964	AKTS_00239753	AKTS_00239757	Email from Mike McLain to Mike McLain and Dave Hodge dated 11/8/18 re: Conversation with Mike McLain
965	N/A	N/A	Qorvo’s Third Supplemental Response to Defendants’ First Set of Interrogatories (Nos. 1-9) dated 8/16/2023 (Attorneys’ Eyes Only)
966	N/A	N/A	Qorvo’s Objections and Responses to Akoustis’ Second Set of Request for Admissions (Nos. 251-300) dated 10/30/23 (Confidential-Attorney’s Eyes Only)
967	N/A	N/A	Qorvo’s Objections and Responses to Akoustis’ Sixth Set of Interrogatories (Nos. 17-19) dated 10/30/23 (Confidential, Attorneys’ Eyes Only)

Dx	Beginning Bates	End Bates	Description
968	AKTS_00036779	AKTS_00036789	Email from Dave Aichele to Dave Hodge dated 9/24/2017 re: 5.2GHz S-Part
969	AKTS_00421215	AKTS_00421240	Germany: DE 102 41 425 A1
970	AKTS_00421241	AKTS_00421487	WIPO: WO 02/093740 A1
971	AKTS_00014596	AKTS_00014614	WIPO: WO 99/10938
972	AKTS_00014615	AKTS_00014635	WIPO: WO 02/23720
973	AKTS_00014583	AKTS_00014595	U.S. Patent: Pat. No. 5,873,154
974	AKTS_00014518	AKTS_00014560	U.S. Patent: Pat. No. 6,081,171
975			P. Osbond et al., Influence of ZnO and Electrode Thickness on the Performance of Thin Film Bulk Acoustic Wave Resonators, vol. 2, 1999 IEEE Ultrasonics Symposium 911 (1999)
976	AKTS_00014561	AKTS_00014582	U.S. Patent: Pat. No. 5,760,663
977	AKTS_00014757	AKTS_00014773	U.S. Patent: Pat. No. 7,709,999 B2
978	SKTS_00014729	AKTS_00014756	U.S. Patent Application: U.S. Application Publication No. US 2014/0159548 A1
979	AKTS_00014642	AKTS_00014663	U.S. Patent: Pat. No. 6,974,211
980	AKTS_000014801	AKTS_00014806	F. Shen et al., Energy Trapping in Mesa-Shaped Quartz Crystal Microbalance, Sensors 2002 IEEE 1154 (2000)
981			Declaration and Assignment of Patent Application
982			U.S. Patent: Pat. No. 11,689,196 B2

DX	Beginning Bates	End Bates	Description
983	AKTS_00089500	AKTS_00089505	Email from Dave Hodge to Annia Shen dated 10/23/19 re: IEEE article
984	AKTS_00134057	AKTS_00134065	Email from Dave Aichele to Annia Shen and Rohan Houlden dated 1/10/19 re: Quorvo 88136 2.4GHz BP BAW filter
985	QORVO_00643138	QORVO_00643215	Email from Eid Alsabbagh to Jerry Gray and Bob Baeten dated 8/7/15 re: 5GHz 5V Low Cost FEM MRD
986	AKTS_00053077	AKTS_00053078	Email from Dave Aichele to Rohan Houlden dated 3/17/20 re: Taisaw
987	AKTS_00053079	AKTS_00053085	Qorvo QPQ1298; B41 Sub-Band 160 MHz BAW Filter
988	AKTS_00807368	AKTS_00807443	USPTO Patent Number: 10,355,659 Issue date: 7/16/19 titled "Piezoelectric Acoustic Resonator Manufactured With Piezoelectric Thin Film Transfer Process"
989	AKTS_01211078	AKTS_01211081	Article - <i>BAW Filters for 5G Bands</i>
990	AKTS_00813658	AKTS_00813660	Akoustis Compensation Study